STIELTJES REGIMES OF PÓLYA–SCHUR MULTIPLIERS AND STRICT DE BRUIJN–NEWMAN BOUNDS FOR THE DAVENPORT–HEILBRONN FUNCTION

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Abstract. We classify the action of real entire Pólya–Schur multipliers $G(-\partial_z^2)$ on the completed Davenport–Heilbronn function $\Xi_{\mathrm{DH}}(z)$ and establish strict unconditional upper and lower bounds for its de Bruijn–Newman constant Λ_{DH}. Expressing the asymptotic multiplier shift through the Stieltjes logarithmic derivative $S_G(u) = \frac{\mathrm{d}}{\mathrm{d}u}\log G(u^2)$, we partition Laguerre–Pólya multipliers into three regimes governed by the Bohr–Kronecker supremum σ^* of $\mathcal{D}_{\mathrm{DH}}(s)$, construct lacunary canonical products showing that upper order $\rho(G) \in (1/2, 1)$ fails without root-gap regularity, and certify in interval ball arithmetic that Pólya's hyperbolic sinc multiplier strictly separates global real-rootedness from the infinite-height Kronecker threshold. Combining a shifted Gauss–Weierstrass contour analysis and equilateral-hyperbola contraction with interval-certified off-real heat-flow witnesses and Platt–Trudgian's bound $\Lambda_\zeta \le 1/5$, we prove unconditionally that $\Lambda_\zeta \le 1/5 < 1153/5000 < \Lambda_{\mathrm{DH}} < 2(\sigma^* - 1/2)^2$.

1. Introduction

Let χ_5 denote the odd primitive Dirichlet character modulo 5 given by $\chi_5(1) = 1$, $\chi_5(2) = \mathrm{i}$, $\chi_5(3) = -\mathrm{i}$, $\chi_5(4) = -1$, and $\chi_5(5) = 0$, and consider the self-dual Dirichlet combination introduced by Davenport and Heilbronn [7] (see Titchmarsh [30, §10.25]):

$$\kappa := \frac{\sqrt{10 - 2\sqrt{5}} - 2}{\sqrt{5} - 1}, \qquad \mathcal{D}_{\mathrm{DH}}(s) := \frac{1 - \mathrm{i}\kappa}{2} L(s, \chi_5) + \frac{1 + \mathrm{i}\kappa}{2} L(s, \overline{\chi_5}) = \sum_{n=1}^{\infty} a_{\mathrm{DH}}(n)\, n^{-s}. \tag{1.1}$$

In the canonical normalization of Riemann [26], Pólya [23, 24], de Bruijn [8], and Newman [21], the completed entire function and its theta-kernel cosine representation take the form

$$\begin{aligned} \xi_{\mathrm{DH}}(s) &:= \left(\frac{5}{\pi}\right)^{(s+1)/2} \Gamma\left(\frac{s+1}{2}\right) \mathcal{D}_{\mathrm{DH}}(s), \\ \Xi_{\mathrm{DH}}(z) &:= \xi_{\mathrm{DH}}\left(\tfrac{1}{2} + \tfrac{\mathrm{i}z}{2}\right) = 2\int_0^\infty \Phi_{\mathrm{DH}}(u)\cos(zu)\,\mathrm{d}u, \end{aligned} \tag{1.2}$$

where $\Phi_{\mathrm{DH}}(u) := 4\mathrm{e}^{3u}\sum_{n=1}^{\infty} n\, a_{\mathrm{DH}}(n)\, \mathrm{e}^{-\pi n^2 \mathrm{e}^{4u}/5}$ satisfies $\Phi_{\mathrm{DH}}(-u) = \Phi_{\mathrm{DH}}(u)$, encoding $\xi_{\mathrm{DH}}(s) = \xi_{\mathrm{DH}}(1-s)$ and $\Xi_{\mathrm{DH}}(z) = \Xi_{\mathrm{DH}}(-z) = \overline{\Xi_{\mathrm{DH}}(\bar{z})}$. Whereas Riemann's $\Xi(z)$ links critical-line zero geometry to classical criteria of Jensen [12], Speiser [28], Hudson [11], Csordas–Smith–Varga [6], and Lagarias [16] and satisfies $0 \le \Lambda_\zeta \le 1/5$ [27, 25, 22], $\mathcal{D}_{\mathrm{DH}}(s)$ lacks an Euler product and possesses infinitely many zeros both in $\mathrm{Re}(s) > 1$ and off the critical line inside $0 < \mathrm{Re}(s) < 1$ [7, 29, 1, 4]. Bombieri and Ghosh [4] proved that $\sigma^* := \sup\{\mathrm{Re}\,\rho : \mathcal{D}_{\mathrm{DH}}(\rho) = 0\}$ is determined via Kronecker's theorem by $\sum_{p\equiv 2,3 \pmod 5} \arctan(p^{-\sigma^*}) = \frac{\pi}{2} - \arctan\kappa$. Dobner [9] proved that every L-function in the extended Selberg class admits a finite de Bruijn–Newman constant satisfying $\Lambda_F \ge 0$; Ki, Kim, and Lee [15] established the strict inequality $\Lambda_\zeta < 1/2$, while recent preprints have studied Dirichlet L-function upper

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bounds [10] and heuristic low-height root tracking for Ξ_{DH} [20]. Our main theorem classifies Pólya–Schur multipliers on Ξ_{DH} and establishes strict certified bounds on Λ_{DH}.

Theorem 1.1 (Main Theorem). *Let $\mathcal{G}_0$ denote the class of real entire Pólya–Schur multipliers $G(w) = \mathrm{e}^{\lambda w}\prod_{j=1}^{\infty}(1+w/w_j)$ with $\lambda \geq 0$, $0 < w_1 \leq w_2 \leq \cdots$, and $\sum_{j=1}^{\infty} w_j^{-1} < \infty$, acting via $G(-\partial_z^2)\Xi_{\mathrm{DH}}(z) = 2\int_0^{\infty} G(u^2)\Phi_{\mathrm{DH}}(u)\cos(zu)\,\mathrm{d}u$. Define the Stieltjes logarithmic derivative $S_G(u) := \frac{\mathrm{d}}{\mathrm{d}u}\log G(u^2) = 4\lambda u + \sum_{j=1}^{\infty}\frac{2u}{w_j+u^2}$ $(u>0)$, and let $N_{\mathrm{off}}(F;T)$ count the non-real zeros of $F(z)$ in $|\operatorname{Re} z| \leq 2T$ with multiplicity.*

(i) Three-regime Stieltjes classification and lacunary obstruction.
   (a) *If $\lambda = 0$ and $\sum_{j=1}^{\infty} w_j^{-1/2} < \infty$ (including every polynomial and every multiplier of upper order $\rho(G) < 1/2$), then $\lim_{u\to\infty} S_G(u) = 0$ and $N_{\mathrm{off}}(G(-\partial_z^2)\Xi_{\mathrm{DH}};T) \gg_G T$ inside $|\operatorname{Im} z| > 1$. In particular, $(1-\beta\partial_z^2)\Xi_{\mathrm{DH}}(z)$ has infinitely many non-real zeros for every $\beta > 0$.*
   (b) *If $\lambda = 0$ and $\lim_{u\to\infty} S_G(u) = \tau \in (0,\infty)$, then $N_{\mathrm{off}}(G(-\partial_z^2)\Xi_{\mathrm{DH}};T) \gg_G T$ for every $0 < \tau < 2\sigma^* - 1$, whereas $N_{\mathrm{off}}(G(-\partial_z^2)\Xi_{\mathrm{DH}};\infty) < \infty$ for every $\tau > 2\sigma^* - 1$.*
   (c) *Every $G \in \mathcal{G}_0$ with $\lambda > 0$ or lower order $\underline{\rho}(G) > 1/2$ satisfies $\lim_{u\to\infty} S_G(u) = +\infty$. Moreover, $N_{\mathrm{off}}(G(-\partial_z^2)\Xi_{\mathrm{DH}};\infty) < \infty$ whenever $\lambda > 0$, or whenever $\lambda = 0$ and $w_{j+1} - w_j = o(\sqrt{w_j})$ as $j \to \infty$ (in particular for $w_j \sim c\,j^{1/\rho}$ with $\rho \in (1/2,1)$). Conversely, for every $\rho \in (1/2,1)$ there exists a lacunary product $G \in \mathcal{G}_0$ of upper order $\rho(G) = \rho$ and lower order $\underline{\rho}(G) = 0$ such that $\liminf_{u\to\infty} S_G(u) = 0$ and $G(-\partial_z^2)\Xi_{\mathrm{DH}}(z)$ has infinitely many non-real zeros in $|\operatorname{Im} z| > 1$.*

(ii) Strict separation between asymptotic type and global real-rootedness. *Comparing $G_{\cosh,h}(w) := \cosh(2h\sqrt{w})$ and Pólya's hyperbolic sinc multiplier $G_{\mathrm{sinc},h}(w) := \frac{\sinh(2h\sqrt{w})}{2h\sqrt{w}}$, both having $\lim_{u\to\infty} S_G(u) = 2h$, the global thresholds $h^*_{\mathrm{global}}(G_\bullet;\Xi_{\mathrm{DH}}) := \inf\{h>0 : G_{\bullet,h}(-\partial_z^2)\Xi_{\mathrm{DH}}$ has only real zeros$\}$ satisfy*

$$h^*_{\mathrm{global}}(G_{\cosh};\Xi_{\mathrm{DH}}) = \sigma^* - \tfrac{1}{2} < \frac{311}{500} < h^*_{\mathrm{global}}(G_{\mathrm{sinc}};\Xi_{\mathrm{DH}})\,. \tag{1.3}$$

(iii) Strict unconditional bounds and separation for Λ_{DH}. *Under backward heat evolution $H_t^{\mathrm{DH}}(z) := \mathrm{e}^{-t\partial_z^2}\Xi_{\mathrm{DH}}(z)$, the de Bruijn–Newman constant $\Lambda_{\mathrm{DH}} := \inf\{t\in\mathbb{R} : H_t^{\mathrm{DH}}$ has only real zeros$\}$ satisfies*

$$\Lambda_\zeta \leq \frac{1}{5} < \frac{1153}{5000} < \Lambda_{\mathrm{DH}} < 2\Big(\sigma^* - \tfrac{1}{2}\Big)^2. \tag{1.4}$$

The canonical threshold constants in Theorem 1.1 are certified in ball arithmetic to satisfy

$$\sigma^* = 1.120362498\ldots, \quad h^*_\infty := \sigma^* - \tfrac{1}{2} = 0.620362498\ldots, \quad 2(h^*_\infty)^2 = 0.769699258\ldots \tag{1.5}$$

1.1. **Proof overview.** Our proof of Theorem 1.1 combines three quantitative and structural ingredients:

(1) *Conjugate-diagram Borel–Laplace representation and Rolle–Hurwitz subproduct monotonicity (Section 2):* Karamata's Tauberian theorem on $S_G(u) \to \tau_0 \in [0,\infty)$ determines the conjugate indicator diagram $[-\frac{\tau_0}{2},\frac{\tau_0}{2}]$ of $G(\frac{1}{4}\zeta^2)$, transferring $G(-\partial_z^2)\Xi_{\mathrm{DH}}$ to a compact stadium contour in the s-plane where Kronecker Diophantine alignment in $\operatorname{Re}(s) > 1$ governs $0 \leq \tau_0 < 2\sigma^* - 1$ and Cauchy–Rouché disks govern $\tau_0 > 2\sigma^* - 1$; greedy subproduct extraction on disjoint intervals (t_m, t_{m+1}) together with Rolle–Hurwitz non-real zero monotonicity extends supercriticality to Regime III.

(2) *Interval-continuum certification of uniform strip-averaging lag (Section 3):* While $G_{\cosh,h}(-\partial_z^2)\Xi_{\mathrm{DH}}$ becomes globally real-rooted right at $h^*_\infty = \sigma^* - \frac{1}{2}$ via the Hermite–Biehler theorem, $G_{\mathrm{sinc},h}(-\partial_z^2)\Xi_{\mathrm{DH}}(z) = \frac{1}{2h}\int_{-h}^{h}\xi_{\mathrm{DH}}(\frac{1}{2}+u+\frac{\mathrm{i}z}{2})\,\mathrm{d}u$ weights interior ordinates where $\Gamma(\frac{s+1}{2})$ grows exponentially slower than at $+h$; a 64-ball parametric Interval–Newton cover across $h \in [\frac{310}{500},\frac{311}{500}] \supset [\sigma^* - \frac{1}{2},\frac{311}{500}]$ near ρ_{2442} eliminates any semigroup assumption and certifies $h^*_{\mathrm{global}}(G_{\mathrm{sinc}};\Xi_{\mathrm{DH}}) > \frac{311}{500}$.

(3) *Shifted Gauss–Weierstrass folding, strict hyperbola contraction, and Interval–Newton heat witnesses (Sections 4–5):* Folding $H_t^{\mathrm{DH}}(2\gamma - 2\mathrm{i}\delta) = A_t(\gamma,\delta) + \overline{A_t(\gamma,-\delta)}$ on $[0,\infty)$, shifting $v \mapsto u + \mathrm{i}\frac{\pi t}{8}$ to cancel Stirling's linear phase identically, and applying the boundary identity $\partial_\gamma A_t = \mathrm{i}\partial_\delta A_t - \frac{\mathrm{i}}{\sqrt{\pi t}}\mathrm{e}^{-\delta^2/t}\xi_{\mathrm{DH}}(\frac{1}{2}+\mathrm{i}\gamma)$ yields a closed-form height $\gamma_0(t)$ above which all zeros of H_t^{DH} are real and simple; de Bruijn's equilateral-hyperbola contraction on $|\operatorname{Re} z| \le 2\gamma_0(t_1)$ yields $\Lambda_{\mathrm{DH}} < 2(\sigma^* - \frac{1}{2})^2$, while ball-arithmetic Interval–Newton contractions at $t = 1/5$ and $t = 1153/5000$ near ρ_{240} and ρ_{531} certify $\Lambda_{\mathrm{DH}} > 1153/5000 > 1/5 \ge \Lambda_\zeta$.

Combining Ingredient (1) with an explicit super-exponentially gapped canonical product gives Theorem 1.1(i); Ingredient (2) establishes (1.3); and Ingredient (3) yields both strict inequalities in (1.4).

1.2. **Notation and conventions.** Throughout, $s = \sigma + \mathrm{i}\gamma$ ($\sigma,\gamma \in \mathbb{R}$) denotes the Dirichlet variable and $z = x - 2\mathrm{i}\delta = 2\gamma - 2\mathrm{i}\delta$ ($\delta = \sigma - \frac{1}{2}$) denotes the spectral variable in (1.2), mapping $\sigma = \frac{1}{2}$ to $z \in \mathbb{R}$ and $|\sigma - \frac{1}{2}| \le h_\infty^*$ to $|\operatorname{Im} z| \le 2h_\infty^*$. Under $\partial_t H_t = -\partial_z^2 H_t$, Λ matches the convention of de Bruijn [8], Newman [21], Rodgers–Tao [27], Polymath [25], and Platt–Trudgian [22]; in the rescaled variable $w = z/2$ with $\widetilde{\Xi}(w) = \xi(\frac{1}{2}+\mathrm{i}w)$, one has $\widetilde{\Lambda} = \Lambda/4$. We abbreviate $\Gamma_{\mathrm{DH}}(s) := (5/\pi)^{(s+1)/2}\Gamma(\frac{s+1}{2})$. Estimates $f \ll_\alpha g$ or $f = \mathcal{O}_\alpha(g)$ denote $|f| \le C(\alpha)|g|$ with $C(\alpha)$ depending only on α. Closed interval balls $[m \pm r]$ denote rigorous enclosures $\{w : |w - m| \le r\}$ computed in `FLINT/Arb` [13].

2. Stieltjes Shift Representation and Pólya–Schur Regimes

Because $\sum_{j=1}^\infty w_j^{-1} < \infty$ for every $G \in \mathcal{G}_0$, Weierstrass's inequality $|G(u^2)| \le \mathrm{e}^{(\lambda + \sum w_j^{-1})|u|^2}$ is dominated on every strip $|\operatorname{Im} u| \le \frac{\pi}{8} - \varepsilon$ by $|\Phi_{\mathrm{DH}}(u)| \ll \mathrm{e}^{-(\pi/6)\mathrm{e}^{4\operatorname{Re} u}}$ as $\operatorname{Re} u \to +\infty$. Consequently, $G(-\partial_z^2)\Xi_{\mathrm{DH}}(z) = 2\int_0^\infty G(u^2)\Phi_{\mathrm{DH}}(u)\cos(zu)\,\mathrm{d}u$ defines a real-even entire function of order 1 satisfying $G(-\partial_z^2)\Xi_{\mathrm{DH}} \not\equiv 0$ (by Fourier cosine uniqueness, since $G(u^2) > 0$ on $[0,\infty)$ and $\Phi_{\mathrm{DH}} \not\equiv 0$), and for any factorization $G = G^{(1)}G^{(2)}$ in $\mathcal{G}_0$, the finite truncations $G_N^{(2)}(w) := \mathrm{e}^{\lambda_2 w}\prod_{j=1}^N(1 + w/w_{2,j})$ satisfy $G_N^{(2)}(-\partial_z^2)[G^{(1)}(-\partial_z^2)\Xi_{\mathrm{DH}}] \to G(-\partial_z^2)\Xi_{\mathrm{DH}}$ locally uniformly on $\mathbb{C}$.

For the Gaussian heat multiplier $G_t(w) = \mathrm{e}^{tw}$ ($t > 0$), Fourier convolution gives $H_t^{\mathrm{DH}}(z) = (4\pi t)^{-1/2}\int_{-\infty}^\infty \mathrm{e}^{-w^2/(4t)}\Xi_{\mathrm{DH}}(z - \mathrm{i}w)\,\mathrm{d}w$. Substituting $z = 2\gamma - 2\mathrm{i}\delta$ and $v = \delta + w/2$, splitting $\mathbb{R} = [0,\infty)\cup(-\infty,0)$, and invoking $\xi_{\mathrm{DH}}(\frac{1}{2} - v + \mathrm{i}\gamma) = \overline{\xi_{\mathrm{DH}}(\frac{1}{2} + v + \mathrm{i}\gamma)}$ for real γ, v yields our central half-line folding and boundary-derivative identity:

$$\boxed{\begin{aligned} H_t^{\mathrm{DH}}(2\gamma - 2\mathrm{i}\delta) &= A_t(\gamma,\delta) + \overline{A_t(\gamma,-\delta)}, \\ A_t(\gamma,\delta) &:= \frac{1}{\sqrt{\pi t}}\int_0^\infty \mathrm{e}^{-(v-\delta)^2/t}\,\xi_{\mathrm{DH}}\Big(\tfrac{1}{2} + v + \mathrm{i}\gamma\Big)\,\mathrm{d}v, \\ \partial_\gamma A_t(\gamma,\delta) &= \mathrm{i}\,\partial_\delta A_t(\gamma,\delta) - \frac{\mathrm{i}}{\sqrt{\pi t}}\,\mathrm{e}^{-\delta^2/t}\,\xi_{\mathrm{DH}}\Big(\tfrac{1}{2} + \mathrm{i}\gamma\Big). \end{aligned}} \tag{2.1}$$

Here the second line follows upon integrating $\partial_\delta(\mathrm{e}^{-(v-\delta)^2/t}) = -\partial_v(\mathrm{e}^{-(v-\delta)^2/t})$ by parts on $[0,\infty)$, justified pre-move by the super-exponential Gaussian decay $\mathrm{e}^{-(v-\delta)^2/t}$ against $|\xi_{\mathrm{DH}}(\frac{1}{2} + v + \mathrm{i}\gamma)| \ll_\gamma (1+v)^C\mathrm{e}^{\frac{v}{2}\log(v+2)}$ as $v \to +\infty$.

We record two classical preservation properties of Laguerre–Pólya differential operators before proving the three-regime classification.

Lemma 2.1 (Pólya [23, 24], Benz [2]; cf. Lagarias–Suzuki [17])**.** *Let $F(z) \not\equiv 0$ be a real-even entire function of genus 0 or 1 whose zeros all satisfy $|\operatorname{Im} z| \le 2h_0$ ($h_0 > 0$).*

(i) *For every $G \in \mathcal{G}_0$ such that $G_N(-\partial_z^2)F \to G(-\partial_z^2)F \not\equiv 0$ locally uniformly on $\mathbb{C}$, all zeros of $G(-\partial_z^2)F(z)$ lie in the strip $|\operatorname{Im} z| \le 2h_0$.*

(ii) *For* $G_{\cosh,h}(w)=\cosh(2h\sqrt{w})$*, all zeros of* $G_{\cosh,h}(-\partial_z^2)F(z)=\frac{1}{2}[F(z-2\mathrm{i}h)+F(z+2\mathrm{i}h)]$ *are real whenever* $h\geq h_0$*, and real and simple whenever* $h>h_0$*.*

Proof. For (i), let $H(z)=c\,z^m\mathrm{e}^{az}\prod_n(1-z/\zeta_n)\mathrm{e}^{\mu_n z/\zeta_n}$ ($c\in\mathbb{R}\setminus\{0\}$, $a\in\mathbb{R}$, $\mu_n\in\{0,1\}$, $\sum_{\zeta_n\neq 0}|\zeta_n|^{-2}<\infty$) be any real entire function of genus 0 or 1 whose zeros satisfy $|\operatorname{Im}\zeta_n|\leq 2h_0$. Grouping non-real zeros into conjugate pairs $\{\zeta_n,\overline{\zeta_n}\}$ cancels $\operatorname{Im}(\mu_n/\zeta_n+\mu_n/\overline{\zeta_n})=0$, so taking imaginary parts of $H'(z_0)/H(z_0)$ gives the absolutely convergent sum $\operatorname{Im}(H'(z_0)/H(z_0))=-\frac{m\operatorname{Im}z_0}{|z_0|^2}-\sum_n\frac{\operatorname{Im}z_0-\operatorname{Im}\zeta_n}{|z_0-\zeta_n|^2}<0$ for every $\operatorname{Im}z_0>2h_0$, whence $(1\pm\alpha\partial_z)H(z_0)\neq 0$ for all $\alpha\in\mathbb{R}$. Iterating across the linear factors of $G_N(-\partial_z^2)F$ (with de Bruijn [8, Thm. 13] for $\mathrm{e}^{-\lambda\partial_z^2}$) and passing to the limit via Hurwitz's theorem gives (i). For (ii), all zeros of $\omega(z):=F(z+2\mathrm{i}h)$ lie in $\operatorname{Im}z\leq 2(h_0-h)\leq 0$, so Jensen's factorwise modulus inequality gives $|\omega(z)|>|\overline{\omega(\overline{z})}|=|F(z-2\mathrm{i}h)|$ on $\operatorname{Im}z>0$, and the Hermite–Biehler theorem yields simplicity on $\mathbb{R}$ when $h>h_0$. □

Lemma 2.2 (Rolle–Hurwitz Non-Real Zero Monotonicity; cf. Pólya [24], Craven–Csordas—Smith [5])**.** *Let* $F(z)$ *be a real entire function of order at most* 1 *with* $N_{\mathrm{off}}(F;\infty)=2M<\infty$*. Then for every* $G\in\mathcal{G}_0$ *such that* $G_N(-\partial_z^2)F\to G(-\partial_z^2)F\not\equiv 0$ *locally uniformly on* $\mathbb{C}$*, one has* $N_{\mathrm{off}}(G(-\partial_z^2)F;\infty)\leq 2M$*.*

Proof. Factor $F=P_{2M}E$, where $\deg P_{2M}=2M$ and E lies in the Laguerre–Pólya class of order at most 1. Approximating E locally uniformly by real-rooted polynomials Q_K of degree K, the polynomial $R_K:=P_{2M}Q_K$ of degree $2M+K$ has at least K real roots (with multiplicity). For any $\alpha\in\mathbb{R}\setminus\{0\}$, Rolle's theorem applied to $\mathrm{e}^{x/\alpha}R_K(x)$ between consecutive real roots and on the ray where $\mathrm{e}^{x/\alpha}R_K(x)\to 0$ produces at least K real roots of $(1+\alpha\partial_z)R_K$, hence at most $2M$ non-real roots. Letting $K\to\infty$ and $G_N(-\partial_z^2)F\to G(-\partial_z^2)F$ via Hurwitz's theorem preserves $N_{\mathrm{off}}\leq 2M$. □

Remark 2.3. For the single derivative $F\mapsto F'$, the refined asymptotic $N_{\mathrm{off}}(F;T)-N_{\mathrm{off}}(F';T)=2D(F;T)+\mathcal{O}(\log T)$ counting real critical points of F/F' was proved for Ξ by Levinson and Montgomery [19] and for real entire functions of genus 0 or 1 by Ki and Kim [14]; here only the global finiteness bound of Lemma 2.2 is needed.

Theorem 2.4 (Three-Regime Stieltjes Classification)**.** *Under the hypotheses of Theorem 1.1, assertions* (i)(a), (i)(b)*, and the positive half of* (i)(c) *hold for every* $G\in\mathcal{G}_0$*.*

Proof. Set $n_G(t):=\#\{j:w_j\leq t\}$. In (a), $\frac{2u}{w_j+u^2}\leq w_j^{-1/2}$ and $\sum w_j^{-1/2}<\infty$, so dominated convergence gives $\lim_{u\to\infty}S_G(u)=0$. In (c), when $\lambda>0$ we have $S_G(u)\geq 4\lambda u\to+\infty$; when $\lambda=0$ and $\underline{\rho}(G)>\frac{1}{2}$, fix $\alpha\in(\frac{1}{2},\underline{\rho}(G))$. Because $u\mapsto u\,S_G(u)=\sum_{j=1}^{\infty}\frac{2u^2}{w_j+u^2}$ is strictly increasing on $(0,\infty)$, integrating $S_G(y)=(yS_G(y))/y$ over $[1,u]$ gives

$$\log G(u^2)-\log G(1)=\int_1^u S_G(y)\,\mathrm{d}y\leq u\,S_G(u)\log u\implies S_G(u)\geq\frac{u^{2\alpha}-\mathcal{O}(1)}{u\log u}\longrightarrow+\infty \tag{2.2}$$

as $u\to+\infty$, since $2\alpha>1$.

Suppose next that $\lambda=0$ and $\lim_{u\to\infty}S_G(u)=\tau_0\in[0,\infty)$, covering (a) ($\tau_0=0$) and (b) ($\tau_0=\tau>0$). Writing $S_G(u)=\int_0^\infty\frac{2u}{t+u^2}\,\mathrm{d}n_G(t)$, Karamata's Tauberian theorem yields $\lim_{t\to\infty}t^{-1/2}n_G(t)=\frac{\tau_0}{\pi}$. Hence the even canonical product $h_G(\zeta):=G(\frac{1}{4}\zeta^2)=\prod_{j=1}^{\infty}(1+\frac{\zeta^2}{4w_j})$, whose zeros $\pm 2\mathrm{i}\sqrt{w_j}$ lie on the imaginary axis with asymptotic density $\frac{\tau_0}{2\pi}$, has indicator $\limsup_{r\to\infty}r^{-1}\log|h_G(r\mathrm{e}^{\mathrm{i}\theta})|=\frac{\tau_0}{2}|\cos\theta|$ and satisfies $\frac{\mathrm{d}}{\mathrm{d}\zeta}\log h_G(\zeta)=\frac{1}{2}S_G(\frac{1}{2}\zeta)\to\frac{\tau_0}{2}$ uniformly in $|\arg\zeta|\leq\frac{\pi}{4}$ as $|\zeta|\to\infty$ [18, Ch. II, Thm. 2]. By Pólya's conjugate-diagram theorem [3, Thm. 5.4.1], the Borel transform $\mathcal{B}_G(\zeta)$ of $h_G(\zeta)$ is holomorphic on $\mathbb{C}\setminus[-\frac{\tau_0}{2},\frac{\tau_0}{2}]$. Since $-\partial_z^2[\xi_{\mathrm{DH}}(\frac{1}{2}+\frac{\mathrm{i}z}{2})]=\frac{1}{4}\partial_s^2\xi_{\mathrm{DH}}(s(z))$ at $s(z):=\frac{1}{2}+\frac{\mathrm{i}z}{2}$, Cauchy–Borel inversion yields, for

every $\varepsilon > 0$ and stadium contour $\Gamma_{\tau_0,\varepsilon}$ at distance ε from $[-\frac{\tau_0}{2}, \frac{\tau_0}{2}]$,

$$F_G(z) := G(-\partial_z^2)\Xi_{\mathrm{DH}}(z) = \frac{1}{2\pi\mathrm{i}} \oint_{\Gamma_{\tau_0,\varepsilon}} \mathcal{B}_G(\zeta)\, \xi_{\mathrm{DH}}\big(s(z)+\zeta\big)\, \mathrm{d}\zeta, \tag{2.3}$$

justified pre-move by compact support of $\Gamma_{\tau_0,\varepsilon}$ and entirety of ξ_{DH}.

Case 1: Subcritical persistence ($0 \le \tau_0 < 2\sigma^* - 1$). Choose $\beta_0 \in (\max(1, \frac{1}{2} + \frac{\tau_0}{2}), \sigma^*)$ and $\eta_0 \in (0, \frac{1}{8})$ so that $\beta_0 - \frac{\tau_0}{2} > \frac{1}{2} + 3\eta_0$ and $\beta_0 > 1 + 3\eta_0$. By Bombieri and Ghosh [4, Thm. 1], Kronecker's theorem furnishes $\gg T$ simple zeros $s_m = \beta_m + \mathrm{i}\gamma_m$ ($T \le \gamma_m \le 2T$) of $\mathcal{D}_{\mathrm{DH}}(s)$ in disjoint unit-separated strips with $|\beta_m - \beta_0| < \eta_0$ and $\inf_{|w-s_m|=r_0} |\mathcal{D}_{\mathrm{DH}}(w)| \ge c_1 > 0$ for fixed $r_0 \in (0, \min(\eta_0/2, 1/4))$. Set $z_m^{\mathrm{sh}} := 2\gamma_m - 2\mathrm{i}(\beta_m - \frac{1}{2} - \frac{\tau_0}{2})$, so $s(z_m^{\mathrm{sh}}) + \frac{\tau_0}{2} = s_m$ and the pairwise disjoint disks $|z - z_m^{\mathrm{sh}}| \le 2r_0$ lie in $\operatorname{Im} z < -2\eta_0 < 0$ (and in $\operatorname{Im} z < -1$ when $\tau_0 = 0$). Fix $\varepsilon \in (0, \eta_0/8)$ and split $\Gamma_{\tau_0,\varepsilon} = \Gamma_+ \cup \Gamma_-$, where $\Gamma_+ := \{\zeta \in \Gamma_{\tau_0,\varepsilon} : \operatorname{Re}\zeta > \frac{\tau_0}{2} - \eta_0\}$ and $\Gamma_- := \Gamma_{\tau_0,\varepsilon} \setminus \Gamma_+$. Set $\Omega(z) := \frac{1}{2}\log\frac{5(s(z)+1)}{2\pi}$ (principal branch on $\operatorname{Im} s(z) > 0$), which is holomorphic on $|z - z_m^{\mathrm{sh}}| \le 2r_0$ with $|h_G(\Omega(z))| = \gamma_m^{\frac{\tau_0}{4}+o(1)}$. By $\xi_{\mathrm{DH}}(w) = \xi_{\mathrm{DH}}(1-w)$ and Phragmén–Lindelöf convexity on $[-\eta_0, 1+\eta_0]$, we have $|\xi_{\mathrm{DH}}(\sigma+\mathrm{i}\gamma)/\Gamma_{\mathrm{DH}}(\operatorname{Re} s(z) + \mathrm{i}\gamma)| \ll_{\eta_0} \gamma^{\frac{1}{2}\max(\sigma, 1-\sigma, 1+\eta_0) - \frac{1}{2}\operatorname{Re} s(z)}$. On $|z - z_m^{\mathrm{sh}}| \le 2r_0$ and $\zeta \in \Gamma_-$, since $1 - (\beta_m - \tau_0) < \beta_m - 4\eta_0$ and $\beta_m > 1 + 2\eta_0$, we have $\max(\sigma, 1-\sigma, 1+\eta_0) \le \operatorname{Re} s(z) + \frac{\tau_0}{2} - \frac{\eta_0}{2}$, so Γ_- contributes $\mathcal{O}_G(\gamma_m^{-\eta_0/4+o(1)}) = o(1)$ after dividing by $\Gamma_{\mathrm{DH}}(s(z))\, h_G(\Omega(z))$. On Γ_+, $\operatorname{Re}(s(z)+\zeta) > 1 + \frac{\eta_0}{2}$; truncating $\mathcal{D}_{\mathrm{DH}}(s(z)+\zeta)$ at $N_m := \lfloor \gamma_m^{1/4} \rfloor$ before re-closing Γ_+, the Dirichlet tail $n > N_m$ satisfies $\sum_{n>N_m} n^{-\operatorname{Re}(s(z)+\zeta)} \ll_{\eta_0} \gamma_m^{\frac{1}{4}(1-\operatorname{Re} s(z) - \operatorname{Re}\zeta)}$, so across $\operatorname{Re}\zeta \in [\frac{\tau_0}{2} - \eta_0, \frac{\tau_0}{2} + \varepsilon]$ it contributes $\mathcal{O}_G(\gamma_m^{-\frac{3}{8}\eta_0 + \frac{\varepsilon}{4} + o(1)}) = o(1)$. For the finite sum $1 \le n \le N_m$, inserting $\Gamma_{\mathrm{DH}}(s(z)+\zeta)/\Gamma_{\mathrm{DH}}(s(z)) = \mathrm{e}^{\Omega(z)\zeta}(1 + \mathcal{O}_G(\gamma_m^{-1}))$ and noting $\operatorname{Re}(\Omega(z) - \log n) \ge \frac{1}{4}\log\gamma_m - \mathcal{O}(1) > 0$ allows us to re-attach Γ_- at cost $\mathcal{O}_G(\gamma_m^{-\eta_0/2+o(1)}) = o(1)$, yielding uniformly on $|z - z_m^{\mathrm{sh}}| \le 2r_0$ the holomorphic identity

$$\begin{aligned} \frac{F_G(z)}{\Gamma_{\mathrm{DH}}(s(z))\, h_G(\Omega(z))} &= \sum_{n=1}^{N_m} a_{\mathrm{DH}}(n)\, n^{-s(z)}\, \frac{h_G(\Omega(z) - \log n)}{h_G(\Omega(z))} + o(1) \\ &= \mathcal{D}_{\mathrm{DH}}\big(s(z) + \tfrac{\tau_0}{2}\big) + o(1) \end{aligned} \tag{2.4}$$

as $\gamma_m \to \infty$: indeed, integrating $\frac{\mathrm{d}}{\mathrm{d}w}\log h_G(w) = \frac{\tau_0}{2} + o(1)$ from $\Omega(z)$ to $\Omega(z) - \log n$ gives $h_G(\Omega(z) - \log n)/h_G(\Omega(z)) = n^{-\tau_0/2 + \theta_{n,m}(z)}$ with $\max_{1\le n\le N_m} |\theta_{n,m}(z)| \to 0$, and dominated convergence against $n^{-(1+\eta_0/4)}$ gives the second line. Rouché's theorem on $|z - z_m^{\mathrm{sh}}| \le 2r_0$ yields a non-real zero in each disk, so $N_{\mathrm{off}}(F_G; T) \gg_G T$.

Case 2: Supercritical real-rootedness ($\tau_0 > 2\sigma^* - 1$ and Regime III). When $\tau_0 = \tau > 2\sigma^* - 1$, by $F_G(-z) = F_G(z)$, $F_G(\bar z) = \overline{F_G(z)}$, and Lemma 2.1(i) it suffices to consider large $\gamma = \frac{1}{2}\operatorname{Re} z$ and $|\delta| \le h_\infty^*$. Choose $\eta_2 \in (0, \frac{1}{8})$ so $\frac{1}{2} - 3\eta_2 + \frac{\tau}{2} > \sigma^* > 1$, and $\varepsilon \in (0, \eta_2/16)$. Because $\sum_{p\equiv 2,3 \pmod 5} \arctan(p^{-\sigma^*-\eta_2}) < \frac{\pi}{2} - \arctan\kappa$ separates the Euler-product argument range uniformly from $\pm(\frac{\pi}{2} - \arctan\kappa) \pmod\pi$, compactness of the Kronecker torus gives $\inf_{\operatorname{Re} w \ge \sigma^* + \eta_2} |\mathcal{D}_{\mathrm{DH}}(w)| \ge c_3 > 0$. Define on $\gamma \ge \gamma_1$, $\delta \in [-2\eta_2, h_\infty^*]$ the non-vanishing holomorphic function

$$\mathcal{M}_G(z) := \Gamma_{\mathrm{DH}}(s(z))\, h_G(\Omega(z))\, \mathcal{D}_{\mathrm{DH}}\big(s(z) + \tfrac{\tau}{2}\big), \qquad \Omega(z) := \frac{1}{2}\log\frac{5(s(z)+1)}{2\pi}, \tag{2.5}$$

which satisfies $|\mathcal{M}_G(z)| = \gamma^{\frac{1}{4} + \frac{1}{2}\delta + \frac{\tau}{4} + o(1)} \mathrm{e}^{-\frac{\pi}{4}\gamma}$. Truncating $\mathcal{D}_{\mathrm{DH}}$ at $N_\gamma := \lfloor \gamma^{1/4} \rfloor$ on $\Gamma_+ := \{\zeta \in \Gamma_{\tau,\varepsilon} : \operatorname{Re}\zeta \ge \frac{\tau}{2} - \eta_2\}$ exactly as in Case 1 gives $\frac{1}{2\pi\mathrm{i}}\int_{\Gamma_+} \mathcal{B}_G(\zeta)\xi_{\mathrm{DH}}(s(z)+\zeta)\,\mathrm{d}\zeta = \mathcal{M}_G(z)(1 + o(1))$ uniformly for $\delta \in [-\eta_2, h_\infty^*]$. For $\delta \in [\frac{\eta_2}{4}, h_\infty^*]$, Phragmén–Lindelöf bounds $\Gamma_{\tau,\varepsilon} \setminus \Gamma_+$ by $\mathcal{O}_G(\gamma^{-\frac{7}{32}\eta_2 + o(1)})|\mathcal{M}_G(z)| = o(1)\cdot|\mathcal{M}_G(z)|$, whence $F_G(z) \ne 0$ on $\delta \in [\frac{\eta_2}{4}, h_\infty^*]$ (and by conjugation on $[-h_\infty^*, -\frac{\eta_2}{4}]$). On the remaining strip $|\delta| \le \frac{\eta_2}{4}$, since $\mathcal{B}_G(-\zeta) = -\mathcal{B}_G(\zeta)$ and $\xi_{\mathrm{DH}}(1 - s(z) + w) = \xi_{\mathrm{DH}}(\overline{s(\bar z)} + w)$, the reflected cap $\Gamma_- := -\Gamma_+$ contributes $\overline{\mathcal{M}_G(\bar z)}(1 +$

$o(1)$) while $|\operatorname{Re}\zeta| < \frac{\tau}{2} - \eta_2$ is $\mathcal{O}_G(\gamma^{-\frac{3}{8}\eta_2+o(1)})\max(|\mathcal{M}_G(z)|, |\overline{\mathcal{M}_G(\bar z)}|)$, giving $F_G(z) = H_{\mathrm{main}}(z) + o(1)\cdot\max(|\mathcal{M}_G(z)|, |\overline{\mathcal{M}_G(\bar z)}|)$ with holomorphic $H_{\mathrm{main}}(z) := \mathcal{M}_G(z) + \overline{\mathcal{M}_G(\bar z)}$. Since $\frac{\mathrm{d}}{\mathrm{d}z}\log\mathcal{M}_G(z) = -\frac{\pi}{8} + \frac{\mathrm{i}}{4}\log\frac{5\gamma}{2\pi} + \mathcal{O}(1)$, we have $\log|\mathcal{M}_G(z)/\overline{\mathcal{M}_G(\bar z)}| = \delta(\log\frac{5\gamma}{2\pi} + \mathcal{O}(1))$, so $|H_{\mathrm{main}}(z)| \geq (1-\mathrm{e}^{-7})\max(|\mathcal{M}_G(z)|, |\overline{\mathcal{M}_G(\bar z)}|)$ for $\frac{8}{\log\gamma} \leq |\delta| \leq \frac{\eta_2}{4}$ (excluding zeros there) and H_{main} has only simple real zeros r_k, separated by consecutive points $x_k \in \mathbb{R}$ where $\arg\mathcal{M}_G(x_k) = k\pi$. On each symmetric rectangle $\mathcal{R}_k := [x_k, x_{k+1}] \times [-\mathrm{i}\frac{16}{\log\gamma}, \mathrm{i}\frac{16}{\log\gamma}]$, both $\mathcal{M}_G(x_k - 2\mathrm{i}\delta)$ and $\overline{\mathcal{M}_G(x_k + 2\mathrm{i}\delta)}$ share phase $k\pi + \frac{\pi}{4}\delta + \mathcal{O}(1/\log\gamma)$ along the vertical sides, so $|H_{\mathrm{main}}(z)| \geq \frac{1}{2}\max(|\mathcal{M}_G(z)|, |\overline{\mathcal{M}_G(\bar z)}|) > |F_G(z) - H_{\mathrm{main}}(z)|$ on $\partial\mathcal{R}_k$. Rouché's theorem gives a single zero of F_G in each $\mathcal{R}_k$, which by $F_G(\bar z) = \overline{F_G(z)}$ is real and simple; hence $N_{\mathrm{off}}(F_G;\infty) < \infty$.

Finally, in (c), if $\lambda > 0$, write $G(w) = \mathrm{e}^{\lambda w}G_1(w)$ with $G_1 \in \mathcal{G}_0$; Proposition 4.3 gives $N_{\mathrm{off}}(H_\lambda^{\mathrm{DH}};\infty) < \infty$, and Lemma 2.2 (applicable since $G_{1,N}(-\partial_z^2)H_\lambda^{\mathrm{DH}} \to G(-\partial_z^2)\Xi_{\mathrm{DH}} \not\equiv 0$ locally uniformly) yields $N_{\mathrm{off}}(G(-\partial_z^2)\Xi_{\mathrm{DH}};\infty) \leq N_{\mathrm{off}}(H_\lambda^{\mathrm{DH}};\infty) < \infty$. If $\lambda = 0$ and $w_{j+1} - w_j = o(\sqrt{w_j})$, set $\tau_1 := 2\sigma^* > 2\sigma^* - 1$ and $t_m := (\frac{\pi m}{\tau_1})^2$. Since $t_{m+1} - t_m = \frac{2\pi}{\tau_1}\sqrt{t_m} + \frac{\pi^2}{\tau_1^2}$ dominates $o(\sqrt{t_m})$, every open interval (t_m, t_{m+1}) $(m \geq m_0)$ contains a root w_{j_m}. Disjointness of (t_m, t_{m+1}) forces $j_{m_0} < j_{m_0+1} < \cdots$ and $w_{j_m}/t_m \to 1$, so $G_{\mathrm{sub}}(w) := \prod_{m=m_0}^{\infty}(1 + w/w_{j_m})$ satisfies $\lim_{u\to\infty} S_{G_{\mathrm{sub}}}(u) = \tau_1 > 2\sigma^* - 1$, giving $N_{\mathrm{off}}(G_{\mathrm{sub}}(-\partial_z^2)\Xi_{\mathrm{DH}};\infty) < \infty$, and Lemma 2.2 applied to $G/G_{\mathrm{sub}} \in \mathcal{G}_0$ completes the proof. □

Proposition 2.5 (Lacunary Canonical Product Counterexample)**.** *Let $\rho \in (1/2, 1)$, define recursively $R_1 := 16$ and $R_k := 2^{2^{R_{k-1}}}$ for $k \geq 2$, and set $M_k := \lfloor R_k^{2\rho}\rfloor$. Then $G_{\mathrm{lac}}(w) := \prod_{k=1}^{\infty}(1 + w/R_k^2)^{M_k} \in \mathcal{G}_0$ has upper order $\rho(G_{\mathrm{lac}}) = \rho$ and lower order $\underline{\rho}(G_{\mathrm{lac}}) = 0$, satisfies $\lim_{k\to\infty} S_{G_{\mathrm{lac}}}(u_k) = 0$ along $u_k := R_k^{(1-\rho)/2}$, and $G_{\mathrm{lac}}(-\partial_z^2)\Xi_{\mathrm{DH}}(z)$ possesses infinitely many non-real zeros inside $|\operatorname{Im} z| > 1$.*

Proof. Because $\sum_{k=1}^{\infty} M_k R_k^{-2} \leq \sum_{k=1}^{\infty} R_k^{2\rho-2} =: C_{\mathrm{lac}} < \infty$ and $n_{G_{\mathrm{lac}}}(t) \leq \sum_{R_j \leq \sqrt{t}} R_j^{2\rho} \ll t^\rho$ with $n_{G_{\mathrm{lac}}}(R_k^2) \geq M_k \sim (R_k^2)^\rho$, Borel's theorem [3, Thm. 2.6.5] gives $G_{\mathrm{lac}} \in \mathcal{G}_0$ and $\rho(G_{\mathrm{lac}}) = \rho$. Since $R_{k-1} = \log_2\log_2 R_k$ $(k \geq 2)$, the lower-shell degree satisfies $D_{k-1} := \sum_{j=1}^{k-1} M_j \ll (\log\log R_k)^{2\rho}$. At $u_k = R_k^{(1-\rho)/2}$, we have $\log G_{\mathrm{lac}}(u_k^2) \leq D_{k-1}\log(1+u_k^2) + \sum_{j=k}^{\infty} M_j u_k^2 R_j^{-2} \ll (\log\log R_k)^{2\rho}\log R_k + R_k^{\rho-1}$, whence $\underline{\rho}(G_{\mathrm{lac}}) \leq \lim_{k\to\infty} \frac{\log^+\log G_{\mathrm{lac}}(u_k^2)}{\log(u_k^2)} = 0$, and $S_{G_{\mathrm{lac}}}(u_k) \leq \frac{2D_{k-1}}{u_k} + \sum_{j=k}^{\infty}\frac{2M_j u_k}{R_j^2} \ll (\log\log R_k)^{2\rho}R_k^{-(1-\rho)/2} + R_k^{-\frac{3}{2}(1-\rho)} \to 0$.

To exhibit non-real zeros in $|\operatorname{Im} z| > 1$, fix $\eta_0 \in (0, \frac{1}{4}(\sigma^* - 1))$ and $r_0 \in (0, \eta_0/2)$. For each large k, set $T_k := R_k^{(1-\rho)/4}$ and select via [4, Thm. 1] a simple zero $s_k = \beta_k + \mathrm{i}\gamma_k$ of $\mathcal{D}_{\mathrm{DH}}(s)$ with $\gamma_k \in [T_k, 2T_k]$, $\beta_k \in (1 + 2\eta_0, \sigma^*)$, and $\inf_{|w-s_k|=r_0}|\mathcal{D}_{\mathrm{DH}}(w)| \geq c_1 > 0$. Set $z_k := 2\gamma_k - 2\mathrm{i}(\beta_k - \frac{1}{2})$ (so $|z - z_k| \leq 2r_0$ lies in $\operatorname{Im} z < -1$) and factor $G_{\mathrm{lac}} = G_{\leq k-1}G_{\geq k}$. Writing $\Delta_k(z) := (G_{\mathrm{lac}}(-\partial_z^2) - G_{\leq k-1}(-\partial_z^2))\Xi_{\mathrm{DH}}(z) = \int_{-\infty}^{\infty} G_{\leq k-1}(u^2)(G_{\geq k}(u^2) - 1)\Phi_{\mathrm{DH}}(u)\mathrm{e}^{\mathrm{i}zu}\,\mathrm{d}u$ and shifting $u = v + \mathrm{i}\alpha_k$ with $\alpha_k := \frac{\pi}{8} - \frac{1}{\gamma_k}$ (justified by double-exponential decay of Φ_{DH} on $|\operatorname{Im} u| < \frac{\pi}{8}$) pulls out $|\mathrm{e}^{\mathrm{i}z(v+\mathrm{i}\alpha_k)}| \asymp \mathrm{e}^{-\frac{\pi}{4}\gamma_k}\mathrm{e}^{2\delta v}$. By $\Phi_{\mathrm{DH}}(-w + \mathrm{i}\alpha_k) = \overline{\Phi_{\mathrm{DH}}(w + \mathrm{i}\alpha_k)}$ and $\cos(4\alpha_k) = \sin(4/\gamma_k) \geq 2/\gamma_k$, we have $|\Phi_{\mathrm{DH}}(v + \mathrm{i}\alpha_k)| \ll \gamma_k\mathrm{e}^{-|v|}$ for $|v| \leq V_k := \log\gamma_k$ and $\ll \mathrm{e}^{3|v| - \frac{2}{5}\gamma_k^{-1}\mathrm{e}^{4|v|}}$ for $|v| > V_k$. On $|v| \leq V_k$, $|G_{\leq k-1}(u^2)| \leq (2 + V_k^2)^{D_{k-1}} = R_k^{o(1)}$ and $|G_{\geq k}(u^2) - 1| \ll (\log R_k)^2 R_k^{2\rho-2}$, while $|v| > V_k$ is $\mathcal{O}(\mathrm{e}^{-\frac{\pi}{4}\gamma_k - \frac{1}{5}\gamma_k^3})$. Dividing by $|\Gamma_{\mathrm{DH}}(s(z))| \asymp \gamma_k^{\frac{1}{4}+\frac{1}{2}\delta}\mathrm{e}^{-\frac{\pi}{4}\gamma_k} \geq \gamma_k^{1/2}\mathrm{e}^{-\frac{\pi}{4}\gamma_k}$ gives $|\Delta_k(z)/\Gamma_{\mathrm{DH}}(s(z))| \ll R_k^{2\rho-2+\frac{1-\rho}{4}(2\sigma^*-1/2)+o(1)} \leq R_k^{-\frac{3}{2}(1-\rho)+o(1)} \to 0$ since $2\sigma^* - \frac{1}{2} < 2$. Meanwhile, for the polynomial $G_{\leq k-1}(w) = \sum_{m=0}^{D_{k-1}} c_{k,m}w^m$ of degree $D_{k-1} \ll (\log\log\gamma_k)^{2\rho}$, writing $\xi_{\mathrm{DH}}(s+\zeta)/\Gamma_{\mathrm{DH}}(s) = \mathrm{e}^{\Omega(z)\zeta}f_s(\zeta)$ and applying Cauchy's bounds $|f_s^{(j)}(0)|/\Gamma(j+1) \leq C_2(\eta_0/4)^{-j}$ on $|\zeta| \leq \eta_0/4$ together with $G_{\leq k-1}(\frac{1}{4}|\Omega(z)|^2) \leq \mathrm{e}^{\frac{1}{2}C_{\mathrm{lac}}}|G_{\leq k-1}(\frac{1}{4}\Omega(z)^2)|$ and

$|G_{\le k-1}(\frac{1}{4}\Omega(z)^2)| \ge 1$ yields $G_{\le k-1}(-\partial_z^2)\Xi_{\mathrm{DH}}(z)/[\Gamma_{\mathrm{DH}}(s(z))G_{\le k-1}(\frac{1}{4}\Omega(z)^2)] = \mathcal{D}_{\mathrm{DH}}(s(z)) + \mathcal{O}(D_{k-1}/\log\gamma_k)$ on $|z-z_k| = 2r_0$. Rouché's theorem produces a zero in $|z-z_k| < 2r_0 \subset \{\operatorname{Im} z < -1\}$ for every large k. □

3. Separation Between Asymptotic Type and Global Real-Rootedness

Both $G_{\cosh,h}(w) = \cosh(2h\sqrt{w})$ and Pólya's hyperbolic sinc multiplier $G_{\mathrm{sinc},h}(w) = \frac{\sinh(2h\sqrt{w})}{2h\sqrt{w}} = \prod_{j=1}^{\infty}(1+\frac{4h^2w}{\pi^2 j^2})$ lie in Regime II of Theorem 1.1(i) with identical asymptotic shift $\lim_{u\to\infty} S_G(u) = 2h$ and identical infinite-height threshold $h^*_\infty = \sigma^* - \frac{1}{2}$. We first enclose σ^* rigorously and then certify that $G_{\mathrm{sinc},h}$ strictly lags $G_{\cosh,h}$ globally.

Lemma 3.1 (Rigorous Enclosure of σ^* and Global Threshold of $G_{\cosh,h}$)**.** *The unique root $\sigma^* > 1$ of Bombieri and Ghosh's equation $\sum_{p\equiv 2,3 \pmod 5} \arctan(p^{-\sigma^*}) = \frac{\pi}{2} - \arctan\kappa$ satisfies*

$$\begin{aligned} \sigma^* &\in \big[1.12036249818332506180145892226464 \pm 10^{-32}\big], \\ h^*_{\mathrm{global}}(G_{\cosh};\Xi_{\mathrm{DH}}) &= \sigma^* - \tfrac{1}{2} \in \Big(\frac{310}{500}, \frac{311}{500}\Big). \end{aligned} \tag{3.1}$$

Proof. Summing $p < 1000$ directly and expanding $\arctan(p^{-\sigma}) = \sum_{m=0}^{14} \frac{(-1)^m}{2m+1} p^{-(2m+1)\sigma} + \mathcal{O}(10^{-95})$ for $p \ge 1000$ expresses each odd prime power sum via Möbius inversion of $\frac{1}{2}\log[(1-5^{-u})\zeta(u)/L(u,\chi_{5,\mathrm{quad}})]$, where $\chi_{5,\mathrm{quad}}$ is the quadratic character modulo 5. Evaluating at $\sigma_{\mathrm{mid}} \pm 10^{-75}$ in 320-bit `FLINT/Arb` arithmetic certifies opposite signs by the Intermediate Value Theorem, giving (3.1). Since all zeros of Ξ_{DH} lie in $|\operatorname{Im} z| \le 2(\sigma^* - \frac{1}{2})$, Lemma 2.1(ii) and Theorem 1.1(i)(b) yield $h^*_{\mathrm{global}}(G_{\cosh};\Xi_{\mathrm{DH}}) = \sigma^* - \frac{1}{2}$. □

Theorem 3.2 (Global Threshold Separation for $G_{\mathrm{sinc},h}$)**.** *For every $h \in (0, \frac{311}{500}]$, the entire function $G_{\mathrm{sinc},h}(-\partial_z^2)\Xi_{\mathrm{DH}}(z)$ possesses at least one non-real zero; in particular, at exact rational $h = \frac{311}{500} = 0.622 > \sigma^* - \frac{1}{2}$, it possesses a certified simple non-real zero inside the closed ball $B_{\mathrm{sinc}} := \{z \in \mathbb{C} : |\operatorname{Re}(z - z_{\mathrm{sinc}})| \le 10^{-45},\ |\operatorname{Im}(z - z_{\mathrm{sinc}})| \le 10^{-45}\}$ centered at*

$$z_{\mathrm{sinc}} = 4884.886654191757156433333 0 - 0.056004628860858072317639 15\,\mathrm{i}, \tag{3.2}$$

originating from the off-line zero $\rho_{2442} = 0.8977501550\ldots + 2442.5302924685\ldots\,\mathrm{i}$ of $\mathcal{D}_{\mathrm{DH}}(s)$. Consequently, $h^*_{\mathrm{global}}(G_{\mathrm{sinc}};\Xi_{\mathrm{DH}}) > \frac{311}{500} > \sigma^* - \frac{1}{2} = h^*_{\mathrm{global}}(G_{\cosh};\Xi_{\mathrm{DH}})$.

Proof. For $0 < h < \sigma^* - \frac{1}{2}$, Theorem 1.1(i)(b) proves $N_{\mathrm{off}}(G_{\mathrm{sinc},h}(-\partial_z^2)\Xi_{\mathrm{DH}};\infty) = \infty$. Because $G_{\mathrm{sinc},h}$ does not form a multiplicative semigroup in h, concluding $h^*_{\mathrm{global}}(G_{\mathrm{sinc}};\Xi_{\mathrm{DH}}) > \frac{311}{500}$ requires certifying non-real zeros across the entire interval $[\sigma^* - \frac{1}{2}, \frac{311}{500}] \subset [\frac{310}{500}, \frac{311}{500}]$. By Fubini's theorem on $G_{\mathrm{sinc},h}(u^2) = \frac{1}{2h}\int_{-h}^{h} \mathrm{e}^{2uv}\,\mathrm{d}v$, define the rescaled entire function $F_{\mathrm{sinc}}(z,h) := 2h\,C_{\mathrm{ref}}^{-1} G_{\mathrm{sinc},h}(-\partial_z^2)\Xi_{\mathrm{DH}}(z) = \int_{-h}^{h} \tilde{\xi}_{\mathrm{DH}}(s(z)+u)\,\mathrm{d}u$, where $C_{\mathrm{ref}} := \Gamma_{\mathrm{DH}}(\frac{1}{2} + 2442.53\,\mathrm{i})$ and $\tilde{\xi}_{\mathrm{DH}} := \xi_{\mathrm{DH}}/C_{\mathrm{ref}}$. By the Fundamental Theorem of Calculus, both first partial derivatives have quadrature-free closed forms:

$$\begin{aligned} \partial_z F_{\mathrm{sinc}}(z,h) &= \frac{\mathrm{i}}{2}\Big[\tilde{\xi}_{\mathrm{DH}}\big(s(z)+h\big) - \tilde{\xi}_{\mathrm{DH}}\big(s(z)-h\big)\Big], \\ \partial_h F_{\mathrm{sinc}}(z,h) &= \tilde{\xi}_{\mathrm{DH}}\big(s(z)+h\big) + \tilde{\xi}_{\mathrm{DH}}\big(s(z)-h\big). \end{aligned} \tag{3.3}$$

At $h = \frac{311}{500}$ and $z_0 = z_{\mathrm{sinc}}$, integrating the 100-term 384-bit `FLINT/Arb` Taylor series of $\tilde{\xi}_{\mathrm{DH}}(s(z_0)+u)$ termwise and bounding $|R_{100}(u)| \le M_4 \frac{(h/4)^{100}}{1-h/4} < 10^{-72}$ via a 512-arc interval cover of $|w| = 4$ ($M_4 \le 6.58 \times 10^7$; Appendix B) yields $N(B_{\mathrm{sinc}}) := z_0 - F_{\mathrm{sinc}}(z_0, \frac{311}{500})/\partial_z F_{\mathrm{sinc}}(B_{\mathrm{sinc}}, \frac{311}{500}) \subset \operatorname{int}(B_{\mathrm{sinc}})$ with $\operatorname{Im} N(B_{\mathrm{sinc}}) < 0$, proving (3.2). Covering $[\frac{310}{500}, \frac{311}{500}]$ by 64 overlapping interval balls $H_k = [h_k \pm 1.564 \times 10^{-5}]$ and applying parametric Interval–Newton contraction $N(B_k, H_k) := z_k - [F_{\mathrm{sinc}}(z_k, h_k) + (H_k - h_k)\partial_h F_{\mathrm{sinc}}(z_k, H_k)]/\partial_z F_{\mathrm{sinc}}(B_k, H_k)$ on boxes B_k of radius 1.5×10^{-3} certifies $N(B_k, H_k) \subset \operatorname{int}(B_k)$ and $\operatorname{Im} N(B_k, H_k) < 0$ across all 64 slices (and by openness slightly beyond $\frac{311}{500}$), establishing (1.3); see Figure 1. □

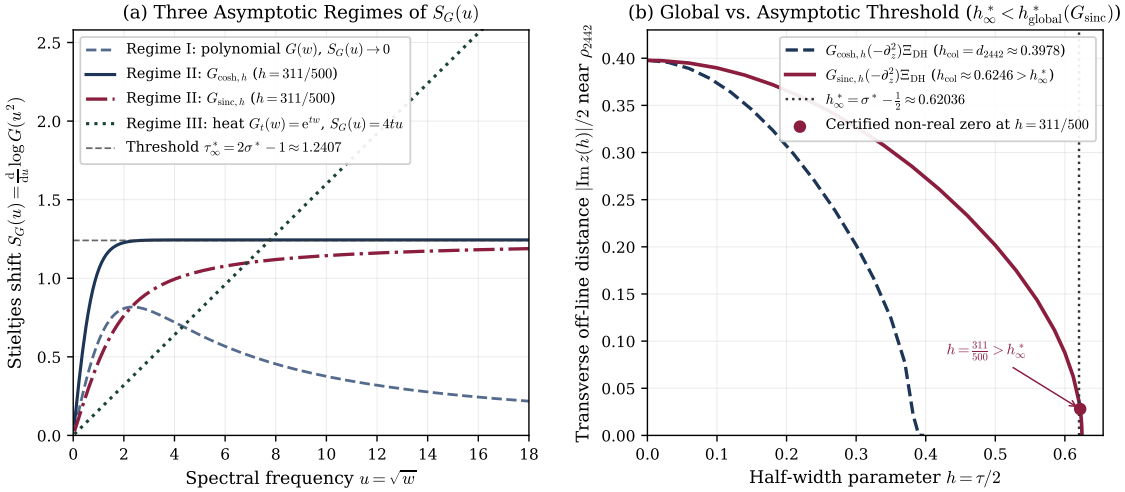


FIGURE 1. (a) Stieltjes logarithmic-derivative profiles $S_G(u)$ across Regimes I, II, and III relative to $\tau_\infty^* = 2\sigma^* - 1$. (b) Transverse off-line ordinate $|\operatorname{Im} z(h)|/2$ near ρ_{2442} under $G_{\cosh,h}$ versus $G_{\mathrm{sinc},h}$, showing certified persistence past $h_\infty^* = \sigma^* - \frac{1}{2}$ across $[\sigma^* - \frac{1}{2}, 311/500]$.

4. Shifted Gauss–Weierstrass Asymptotics and Strict Upper Bound

For $G_t(w) = \mathrm{e}^{tw}$, de Bruijn's hyperbola and strip theorems govern zero contraction under the backward heat semigroup.

Lemma 4.1 (de Bruijn [8, Thms. 12 and 13])**.** *Let $F(z)$ be a real-even entire function of genus 0 or 1 whose zeros all satisfy $|\operatorname{Im} z| \le \Delta_0$.*

(i) *For every $t > 0$, all zeros of $\mathrm{e}^{-t\partial_z^2}F(z)$ satisfy $|\operatorname{Im} z| \le \sqrt{\max(0, \Delta_0^2 - 2t)}$; in particular, $\Lambda_{\mathrm{DH}} \le 2(\sigma^* - \frac{1}{2})^2$.*

(ii) *If $x_* \in \mathbb{R}$ and $\widetilde{\Delta} \ge 0$ are such that every zero z_ρ of F satisfies $(\operatorname{Im} z_\rho)^2 - (\operatorname{Re} z_\rho - x_*)^2 \le \widetilde{\Delta}^2$, then every zero z of $\mathrm{e}^{-t\partial_z^2}F(z)$ on the vertical line $\operatorname{Re} z = x_*$ satisfies $(\operatorname{Im} z)^2 \le \max(0, \widetilde{\Delta}^2 - 2t)$.*

To sharpen Lemma 4.1(i) to a *strict* inequality $\Lambda_{\mathrm{DH}} < 2(\sigma^* - \frac{1}{2})^2$, we prove via the half-line folding (2.1) that for every $t > 0$, $H_t^{\mathrm{DH}}(z)$ has only real, simple zeros above an explicit closed-form height $\gamma_0(t)$, avoiding the incomplete-Gamma boundary-saddle transition required in Riemann–Siegel heat-kernel expansions [25]. We first state the two elementary Dirichlet bounds proved in Appendix A.

Lemma 4.2 (Explicit Abel and Dirichlet-Tail Bounds)**.** *For all $\tau \ge 100$ and $v \ge 0$, $|\mathcal{D}_{\mathrm{DH}}(\frac{1}{2} + v + \mathrm{i}\tau)| \le 4\tau^{1/2}$. Moreover, for all $\tau \in \mathbb{R}$ and $u \ge \frac{5}{4}$, $\sum_{n=2}^{\infty} |a_{\mathrm{DH}}(n)| n^{-1/2-u} < \frac{1}{2}$, so $\operatorname{Re} \mathcal{D}_{\mathrm{DH}}(\frac{1}{2} + u + \mathrm{i}\tau) > \frac{1}{2}$ and $|\mathcal{D}_{\mathrm{DH}}(\frac{1}{2} + u + \mathrm{i}\tau)| < \frac{3}{2}$.*

Proposition 4.3 (Explicit Asymptotic Real-Rootedness and Simplicity for H_t^{DH})**.** *Fix $t > 0$ and set $\Delta := \sigma^* - \frac{1}{2} < \frac{5}{8}$. Define the explicit closed-form cutoff height*

$$\gamma_0(t) := \max\left(400 + 64t^3,\ \frac{2\pi}{5}\exp\left(\frac{4}{t}\left(\Delta + 5 + 6\sqrt{t} + 2t\right)\right)\right). \tag{4.1}$$

Then for all $|\gamma| \ge \gamma_0(t)$, $H_t^{\mathrm{DH}}(2\gamma - 2\mathrm{i}\delta) \ne 0$ for every $\delta \ne 0$, and every real zero of $H_t^{\mathrm{DH}}(x)$ in $|x| \ge 2\gamma_0(t)$ is simple.

Proof. By evenness and Lemma 4.1(i), it suffices to consider $\gamma \ge \gamma_0(t)$ and $|\delta| \le \Delta$. In (2.1), fix any $\eta \in [-\Delta, \Delta]$ and deform the ray $v \in [0, \infty)$ to $\mathcal{V}_\theta \cup \mathcal{C}_\theta$, where $\mathcal{V}_\theta := \{\mathrm{i}y : 0 \le y \le \theta\}$, $\mathcal{C}_\theta := \{u + \mathrm{i}\theta : u \ge 0\}$, and $\theta := \frac{\pi t}{8}$; this contour shift is justified pre-move because ξ_{DH} is

entire and $|\mathrm{e}^{-(u+\mathrm{i}y-\eta)^2/t}\xi_{\mathrm{DH}}(\frac12+u+\mathrm{i}(\gamma+y))| \ll_{\gamma,\theta} \mathrm{e}^{-u^2/(2t)} \to 0$ as $u\to+\infty$ uniformly for $y\in[0,\theta]$. Set $\tilde\gamma := \gamma+\theta > \gamma$ and $K_0(\gamma,\eta) := \mathrm{e}^{\frac{\pi^2 t}{64}+\mathrm{i}\frac{\pi}{4}\eta}\Gamma_{\mathrm{DH}}(\frac12+\mathrm{i}\tilde\gamma) \neq 0$. Along $v=u+\mathrm{i}\theta$, expanding $-(u+\mathrm{i}\theta-\eta)^2/t = -\frac{(u-\eta)^2}{t} + \frac{\pi^2 t}{64} - \mathrm{i}\frac{\pi}{4}(u-\eta)$ cancels the $+\mathrm{i}\frac{\pi}{4}u$ Stirling phase of $\Gamma_{\mathrm{DH}}(\frac12+u+\mathrm{i}\tilde\gamma)/\Gamma_{\mathrm{DH}}(\frac12+\mathrm{i}\tilde\gamma) = (\frac{5\tilde\gamma}{2\pi})^{u/2}\mathrm{e}^{\mathrm{i}\frac{\pi}{4}u+E_{\mathrm{St}}(u,\tilde\gamma)}$ identically, yielding

$$\frac{\sqrt{\pi t}\,\mathrm{e}^{-u_0^2/t+\eta^2/t}}{K_0(\gamma,\eta)}\,A_t(\gamma,\eta) = \int_0^\infty \mathrm{e}^{-(u-u_0)^2/t+E_{\mathrm{St}}(u,\tilde\gamma)}\,\mathcal{D}_{\mathrm{DH}}\Big(\tfrac12+u+\mathrm{i}\tilde\gamma\Big)\,\mathrm{d}u + \mathcal{E}_{\mathcal{V}}(\gamma,\eta), \tag{4.2}$$

where $u_0 = u_0(\gamma,\eta) := \eta + \frac{t}{4}\log\frac{5\tilde\gamma}{2\pi}$. Under (4.1), $u_0 > 5+6\sqrt{t}+2t$ for all $\eta\in[-\Delta,\Delta]$. As proved in Appendix A, Binet's digamma bound gives $|E_{\mathrm{St}}(u,\tilde\gamma)| \le \frac{u^2+6u}{4\tilde\gamma}$ for all $u\ge0$, so $|E_{\mathrm{St}}(u,\tilde\gamma)| < \frac{1}{10}$ on $|u-u_0|\le 3\sqrt{t}$ (where $u>5>\frac54$ and $\mathrm{Re}(\mathrm{e}^{E_{\mathrm{St}}}\mathcal{D}_{\mathrm{DH}}) \ge \mathrm{e}^{-1/10}(\frac12\cos\frac{1}{10} - \frac32\sin\frac{1}{10}) > 0.31$), while the split left tail $[0,\frac54]\cup[\frac54,u_0-3\sqrt{t}]$, vertical segment $\mathcal{V}_\theta$, and right tail $u\ge u_0+3\sqrt{t}$ contribute less than $0.015\sqrt{\pi t}$ in modulus, leaving real part $> 0.29\sqrt{\pi t}$. Inserting $\frac{2}{t}(v-\eta) = (\frac12\log\frac{5\tilde\gamma}{2\pi}+\mathrm{i}\frac{\pi}{4}) + \frac{2}{t}(u-u_0)$ for $\partial_\eta A_t(\gamma,\eta)$ bounds the centered-moment remainder by $\frac{6.7}{\sqrt{t}} < \frac14\log\frac{5\tilde\gamma}{2\pi}$ (Appendix A), showing that $A_t(\gamma,\eta)\neq0$ and

$$\mathrm{Re}\Big(\frac{\partial_\eta A_t(\gamma,\eta)}{A_t(\gamma,\eta)}\Big) \ge \frac14\log\frac{5\tilde\gamma}{2\pi} > 1 \qquad \forall\,\eta\in[-\Delta,\Delta]. \tag{4.3}$$

Consequently, for every $\delta\in(0,\Delta]$, integrating (4.3) over $[-\delta,\delta]$ gives $\log|A_t(\gamma,\delta)| - \log|A_t(\gamma,-\delta)| = \int_{-\delta}^{\delta}\mathrm{Re}(\partial_\eta A_t/A_t)\,\mathrm{d}\eta > 0$, whence $|H_t^{\mathrm{DH}}(2\gamma-2\mathrm{i}\delta)| \ge |A_t(\gamma,\delta)| - |A_t(\gamma,-\delta)| > 0$. Finally, at $\delta=0$, $H_t^{\mathrm{DH}}(2\gamma) = 2|A_t(\gamma,0)|\cos\Theta_t(\gamma)$ with $\Theta_t(\gamma) := \arg A_t(\gamma,0)$; dividing the second identity of (2.1) at $\delta=0$ by $A_t(\gamma,0)$ and applying the explicit boundary ratio bound $< 10^{-2}$ from Appendix A yields

$$\Theta_t'(\gamma) = \mathrm{Re}\Big(\frac{\partial_\delta A_t(\gamma,0)}{A_t(\gamma,0)}\Big) - \mathrm{Re}\left(\frac{\xi_{\mathrm{DH}}(\frac12+\mathrm{i}\gamma)}{\sqrt{\pi t}\,A_t(\gamma,0)}\right) \ge \frac14\log\frac{5\gamma}{2\pi} - 10^{-2} > 1 > 0, \tag{4.4}$$

so $\frac{\mathrm{d}}{\mathrm{d}\gamma}H_t^{\mathrm{DH}}(2\gamma) = \mp 2|A_t(\gamma,0)|\Theta_t'(\gamma) \neq 0$ at every real zero $\cos\Theta_t(\gamma)=0$. □

Theorem 4.4 (Strict Unconditional Upper Bound for Λ_{DH})**.** *The de Bruijn–Newman constant of $\Xi_{\mathrm{DH}}(z)$ satisfies the strict inequality*

$$\Lambda_{\mathrm{DH}} < 2\Big(\sigma^* - \tfrac12\Big)^2 = 0.76969925830451198120\ldots \tag{4.5}$$

Proof. Set $\Delta_0 := 2(\sigma^*-\frac12)$ and fix $t_1 := \frac12 < \frac12\Delta_0^2 = 2(\sigma^*-\frac12)^2$. By Proposition 4.3, all zeros of $H_{t_1}^{\mathrm{DH}}(z)$ in $|\mathrm{Re}\,z| \ge 2\gamma_0(t_1)$ are real, so $H_{t_1}^{\mathrm{DH}}$ possesses at most finitely many non-real zeros, all confined to $|\mathrm{Re}\,z| < 2\gamma_0(t_1)$. If $H_{t_1}^{\mathrm{DH}}$ has no non-real zeros, then $\Lambda_{\mathrm{DH}} \le t_1 < 2(\sigma^*-\frac12)^2$. Otherwise, $Y_1 := \max\{|\mathrm{Im}\,w| : H_{t_1}^{\mathrm{DH}}(w)=0\} > 0$ is attained at some $w_* = x_* + \mathrm{i}y_*$ with $|x_*| < 2\gamma_0(t_1)$.

We verify first that every individual zero $z_\rho = x_\rho + \mathrm{i}y_\rho$ of $\Xi_{\mathrm{DH}}(z)$ satisfies $|y_\rho| < \Delta_0$ strictly. Zeros with $0 \le \mathrm{Re}\,\rho \le 1$ satisfy $|y_\rho| \le 1 < \Delta_0$. For $\mathrm{Re}\,\rho = \sigma > 1$ (and symmetrically $\mathrm{Re}\,\rho < 0$), $\mathcal{D}_{\mathrm{DH}}(\sigma+\mathrm{i}\gamma) = 0$ implies $L(\sigma+\mathrm{i}\gamma,\overline{\chi_5})/L(\sigma+\mathrm{i}\gamma,\chi_5) = \mathrm{e}^{-2\mathrm{i}(\frac{\pi}{2}-\arctan\kappa)}$. In the Euler product, primes $p\equiv1,4 \pmod 5$ cancel identically, while for each $p\equiv2,3\pmod5$ the Möbius factor $\frac{1+w_p}{1-w_p}$ ($w_p := \chi_5(p)p^{-\sigma-\mathrm{i}\gamma}$, $|w_p| = p^{-\sigma}$) maps $|w| = p^{-\sigma}$ onto the circle over $[\frac{1-p^{-\sigma}}{1+p^{-\sigma}}, \frac{1+p^{-\sigma}}{1-p^{-\sigma}}]$, giving $|\arg\frac{1+w_p}{1-w_p}| \le \arcsin(\frac{2p^{-\sigma}}{1+p^{-2\sigma}}) = 2\arctan(p^{-\sigma})$, with equality if and only if $\mathrm{Re}\,w_p = 0$ (i.e., $p^{-\mathrm{i}\gamma} = \pm1$). Since $\frac{\log3}{\log2} \notin \mathbb{Q}$, equality cannot hold simultaneously for $p=2$ and $p=3$ at any $\gamma\in\mathbb{R}$, forcing $\sigma<\sigma^*$ and $|y_\rho|<\Delta_0$.

Now set $R := 2\gamma_0(t_1)+\Delta_0+1$. The finitely many zeros of Ξ_{DH} in $|\mathrm{Re}\,z_\rho| \le R$ satisfy $\Delta_R := \max_{|\mathrm{Re}\,z_\rho|\le R}|\mathrm{Im}\,z_\rho| < \Delta_0$, while every zero with $|\mathrm{Re}\,z_\rho| > R$ satisfies $|\mathrm{Re}\,z_\rho - x_*| > \Delta_0+1 > |\mathrm{Im}\,z_\rho|$, so $(\mathrm{Im}\,z_\rho)^2 - (\mathrm{Re}\,z_\rho - x_*)^2 < 0$. Hence every zero of Ξ_{DH} satisfies $(\mathrm{Im}\,z_\rho)^2 - (\mathrm{Re}\,z_\rho - x_*)^2 \le \Delta_R^2 < \Delta_0^2$. Lemma 4.1(ii) at $\mathrm{Re}\,z = x_*$ yields $Y_1^2 \le \max(0, \Delta_R^2 - 2t_1)$,

and applying Lemma 4.1(i) to $H^{\mathrm{DH}}_{t_1+t_2} = \mathrm{e}^{-t_2\partial_z^2} H^{\mathrm{DH}}_{t_1}$ with $t_2 := \frac{1}{2}Y_1^2$ gives $\Lambda_{\mathrm{DH}} \le t_1 + \frac{1}{2}Y_1^2 \le \max(t_1, \frac{1}{2}\Delta_R^2) < \frac{1}{2}\Delta_0^2 = 2(\sigma^* - \frac{1}{2})^2$. □

5. Interval-Certified Separation $\Lambda_\zeta \le 1/5 < 1153/5000 < \Lambda_{\mathrm{DH}}$

5.1. **Rigorous theta-kernel truncation and Interval–Newton certificates.** To establish unconditional lower bounds on Λ_{DH} free of quadrature aliasing or uncertified root-tracking heuristics, we evaluate $H_t^{\mathrm{DH}}(z) = 2\int_0^\infty \mathrm{e}^{tu^2}\Phi_{\mathrm{DH}}(u)\cos(zu)\,\mathrm{d}u$ and $\partial_z H_t^{\mathrm{DH}}(z)$ directly in arbitrary-precision complex ball arithmetic [13] on a finite interval $[0, U]$ with certified analytical remainder balls.

Lemma 5.1 (Rigorous Theta-Kernel Truncation and Second-Derivative Bounds)**.** *Let $0 \le t \le \frac{1}{4}$, $2 \le U \le \frac{5}{2}$, $N \ge 25$, and $|y| \le 1$, where $y = \operatorname{Im} z$. Approximating $\Phi_{\mathrm{DH}}(u)$ on $[0, U]$ by $\Phi_{\mathrm{DH}}^{(<N)}(u) := 4\mathrm{e}^{3u}\sum_{n=1}^{N-1} n\, a_{\mathrm{DH}}(n)\,\mathrm{e}^{-\pi n^2 \mathrm{e}^{4u}/5}$ incurs a total truncation error bounded for $k \in \{0,1,2\}$ ($\partial_z^k H_t^{\mathrm{DH}}$) by*

$$\left|\partial_z^k H_t^{\mathrm{DH}}(z) - 2\int_0^U \mathrm{e}^{tu^2}\Phi_{\mathrm{DH}}^{(<N)}(u)\,\partial_z^k\cos(zu)\,\mathrm{d}u\right| \le E_1(U,N,k,|y|) + E_2(U), \tag{5.1}$$

where $r_N := \mathrm{e}^{-\pi N/5}$, $E_1(U,N,k,|y|) := 8U^{k+1}\mathrm{e}^{tU^2+(3+|y|)U}\frac{r_N^N(N-(N-1)r_N)}{(1-r_N)^2}$, and $E_2(U) := \frac{40}{\pi}\mathrm{e}^{-4U-\frac{\pi}{10}\mathrm{e}^{4U}}$. Moreover, on any closed box B with $\max_{z\in B}|\operatorname{Im} z| \le y_{\max} \le 1$, one has

$$\sup_{z\in B}\left|\partial_z^2 H_t^{\mathrm{DH}}(z)\right| \le 4N(N-1)U^3\mathrm{e}^{tU^2+3U}\cosh(Uy_{\max}) + E_1(U,N,2,y_{\max}) + E_2(U). \tag{5.2}$$

The proof of Lemma 5.1 is given in Appendix B. Combining Lemma 5.1 with complex Interval–Newton contraction $N(B) := z_0 - H_t^{\mathrm{DH}}(z_0)/[H_t^{\mathrm{DH}\prime}(z_0) + (B - z_0)M_2(B)]$ yields our certified separation theorem.

Theorem 5.2 (Interval-Certified Lower Bounds and Separation from Λ_ζ)**.** *Let $B(z_0, r) := \{z \in \mathbb{C} : |\operatorname{Re}(z - z_0)| \le r,\ |\operatorname{Im}(z - z_0)| \le r\}$.*

(i) *At exact rational $t = \frac{1}{5} = 0.2$, $H_{1/5}^{\mathrm{DH}}(z)$ possesses two independent certified simple non-real zeros in $\operatorname{Im} z < 0$: one inside $B(z_{240}^{(1/5)}, 10^{-90})$ originating from ρ_{240}, and a second inside $B(z_{531}^{(1/5)}, 10^{-193})$ originating from ρ_{531} (Table 1).*

(ii) *At exact rational $t = \frac{1153}{5000} = 0.2306$, $H_{1153/5000}^{\mathrm{DH}}(z)$ possesses a certified simple non-real zero inside $B(z_{240}^{(1153/5000)}, 10^{-94})$ with $\operatorname{Im} z < -3.83 \times 10^{-3} < 0$ (Table 1).*

By Lemma 4.1(i) and Platt and Trudgian's unconditional bound $\Lambda_\zeta \le \frac{1}{5}$ [22], it follows unconditionally that $\Lambda_\zeta \le \frac{1}{5} < \frac{1153}{5000} < \Lambda_{\mathrm{DH}}$.

Proof. Using 640-bit, 704-bit, and 1350-bit `FLINT/Arb` Gauss–Legendre ball quadrature (`acb.integral` [13]) with the analytical remainder enclosures of Lemma 5.1, our verification suite evaluates $H_t^{\mathrm{DH}}(z_0)$, $\partial_z H_t^{\mathrm{DH}}(z_0)$, and (5.2) at each rational parameter t and midpoint z_0 listed in Table 1. In every case, $0 \notin \partial_z H_t^{\mathrm{DH}}(z_0) + (B - z_0)M_2(B)$, $N(B)$ lies strictly inside $\operatorname{int}(B)$, and $\sup_{w\in N(B)}\operatorname{Im} w < 0$. By Brouwer's fixed-point theorem and the Argument Principle, H_t^{DH} has a unique simple zero in B, which is strictly non-real. If $H_{t_0}^{\mathrm{DH}}$ had only real zeros for some $t_0 \le \frac{1153}{5000}$, then Lemma 4.1(i) applied to $H_{1153/5000}^{\mathrm{DH}} = \mathrm{e}^{-(1153/5000-t_0)\partial_z^2} H_{t_0}^{\mathrm{DH}}$ would force all zeros at $t = \frac{1153}{5000}$ to be real, contradicting (ii); openness of simple non-real zeros in t gives the strict inequality $\Lambda_{\mathrm{DH}} > \frac{1153}{5000}$. □

5.2. **Numerical high-height collision survey versus certified theorems.** We strictly distinguish the ball-certified theorems in Table 1 from exploratory numerical computations:

TABLE 1. Summary of `FLINT/Arb` ball-arithmetic enclosures and Interval–Newton contractions ($N(B) \subset \mathrm{int}(B)$) verified by `verify_paper15_debruijn_newman_dh.py`.

Target / Witness	Param.	Certified Midpoint / Enclosure (16 digits shown)	Radius r
Bohr supremum σ^*	IVT	1.1203624981833251	10^{-32}
G_{sinc} (ρ_{2442}, 384-bit)	$h = \frac{311}{500}$	$4884.886654191757 - 0.0560046288608581\,\mathrm{i}$	10^{-45}
G_{sinc} continuum	$[\frac{310}{500}, \frac{311}{500}]$	64 overlapping balls H_k, $\mathrm{Im}\, N(B_k, H_k) < 0$	1.5×10^{-3}
H_t^{DH} (ρ_1, 384-bit)	$t = \frac{22}{125}$	$171.2354942735178 - 0.0317958184889241\,\mathrm{i}$	10^{-48}
H_t^{DH} (ρ_{240}, 640-bit)	$t = \frac{1}{5}$	$480.7066441211221 - 0.2501436731312467\,\mathrm{i}$	10^{-90}
H_t^{DH} (ρ_{531}, 1350-bit)	$t = \frac{1}{5}$	$1062.459584704108 - 0.0505424679933243\,\mathrm{i}$	10^{-193}
H_t^{DH} (ρ_{240}, 704-bit)	$t = \frac{1153}{5000}$	$480.6902350136200 - 0.0038371695655869\,\mathrm{i}$	10^{-94}

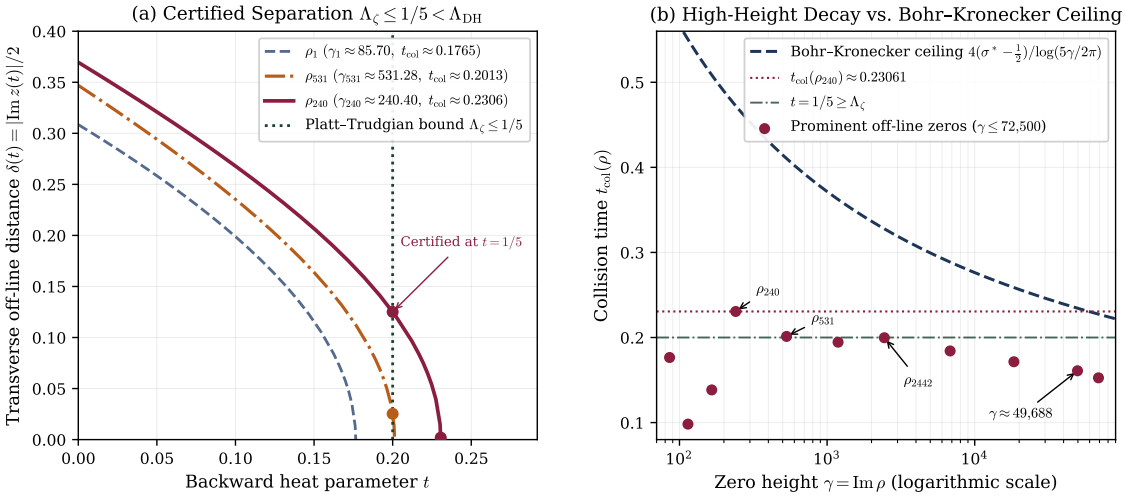


FIGURE 2. (a) Transverse off-line distance $\delta(t) = |\mathrm{Im}\, z(t)|/2$ under backward heat flow for ρ_1, ρ_{240}, and ρ_{531}, showing certified non-real witnesses at $t = 1/5 \ge \Lambda_\zeta$ and $t = 1153/5000 = 0.2306$. (b) Numerical survey of prominent off-line collision times up to $\gamma = 72{,}500$ compared with the asymptotic Bohr–Kronecker ceiling $4(\sigma^* - \frac{1}{2})/\log(\frac{5\gamma}{2\pi})$.

(1) *Oscillation aliasing in non-interval quadrature:* Near the first off-line zero $\rho_1 \approx 0.808517 + 85.699348\,\mathrm{i}$ ($z \approx 171.40 - 0.6170\,\mathrm{i}$), $\cos(zu)$ undergoes roughly 60 oscillations on $[0, 2.25]$. Uncertified adaptive Gauss–Legendre quadrature (`mpmath.quad` at default subdivision) aliases across sign changes and reports a spurious coalescence near $t \approx 0.14194$, whereas the 384-bit `FLINT/Arb` certificate in Table 1 proves $\mathrm{Im}\, z_1(\frac{22}{125}) < 0$ at $t = \frac{22}{125} = 0.176$ (high-precision collision refinement: $t_{\mathrm{col}}(\rho_1) \approx 0.1765053782$).

(2) *High-precision collision coordinates and high-height scan:* Scalar Newton refinement on $(H_t^{\mathrm{DH}}, \partial_z H_t^{\mathrm{DH}}) = (0, 0)$ locates the coalescence of ρ_{240} at $t_{\mathrm{col}}(\rho_{240}) \approx 0.2306073618964546$ ($x_{\mathrm{col}} \approx 480.6902310425$)—matching the uncertified six-digit value 0.230607 noted heuristically in [20]—and that of ρ_{531} at $t_{\mathrm{col}}(\rho_{531}) \approx 0.20127583$. An exploratory scan of off-line zeros of $\mathcal{D}_{\mathrm{DH}}(s)$ up to $\gamma = 72{,}500$ (Figure 2(b)) detects no collision exceeding $t_{\mathrm{col}}(\rho_{240})$: although deeper real parts occur at higher ordinates (such as ρ_{2442} with $\mathrm{Re}\,\rho \approx 0.89775$ and $t_{\mathrm{col}} \approx 0.19973$, or $\gamma \approx 49{,}688.34$ with $\mathrm{Re}\,\rho \approx 0.93187$ and $t_{\mathrm{col}} \approx 0.1609$), Stirling's logarithmic factor $\frac{1}{2}\log\frac{5\gamma}{2\pi}$ accelerates transverse heat

speed according to the asymptotic ceiling $t_{\mathrm{col}}(\rho) \lesssim \frac{4(\sigma^*-1/2)}{\log(5\gamma/(2\pi))} \to 0$ implied by Proposition 4.3. Whether Λ_{DH} coincides with $t_{\mathrm{col}}(\rho_{240})$ remains an open finite-box verification problem.

6. Scope, Limitations, and Open Questions

Theorem 1.1 demonstrates that for self-dual Dirichlet series without an Euler product, Pólya–Schur differential multipliers exhibit two distinct phenomena absent under the Riemann Hypothesis: an infinite-height phase transition governed by the Stieltjes logarithmic derivative $S_G(u)$ relative to $2\sigma^*-1$ (sensitive to root-gap regularity even when $\rho(G) \in (1/2, 1)$), and a strict gap between the infinite-height Kronecker threshold $h^*_\infty = \sigma^* - \frac{1}{2}$ and global real-rootedness for strip-averaging multipliers such as $G_{\mathrm{sinc},h}$. Closing the rigorous interval $\frac{1153}{5000} < \Lambda_{\mathrm{DH}} < 2(\sigma^* - \frac{1}{2})^2$ to arbitrary precision reduces, via Proposition 4.3, to a finite two-dimensional argument-principle box cover up to height $\gamma_0(t)$ once $\gamma_0(t)$ is optimized via stationary-phase Dirichlet sums.

Appendix A. Proofs of Explicit Analytic Constants and Gaussian Tail Bounds

Proof of Lemma 4.2. Set $S(N) := \sum_{n=1}^{N} a_{\mathrm{DH}}(n)$. Because $a_{\mathrm{DH}}(n+5) = a_{\mathrm{DH}}(n)$ and $\sum_{r=1}^{5} a_{\mathrm{DH}}(r) = 1+\kappa-\kappa-1+0 = 0$, checking $N \equiv 1, 2, 3, 4, 0 \pmod 5$ gives $\max_{N\geq 1} |S(N)| = 1 + \kappa < 1.2841$. Fix $\tau \geq 100$, $v \geq 0$, set $\sigma := \frac{1}{2} + v \geq \frac{1}{2}$, and choose $M := \lfloor \tau \rfloor \geq 100$. Splitting $\mathcal{D}_{\mathrm{DH}}(\sigma + \mathrm{i}\tau) = \sum_{n=1}^{M} a_{\mathrm{DH}}(n) n^{-\sigma-\mathrm{i}\tau} + \sum_{n=M+1}^{\infty} a_{\mathrm{DH}}(n) n^{-\sigma-\mathrm{i}\tau}$, the prefix satisfies

$$\begin{aligned}\left|\sum_{n=1}^{M} a_{\mathrm{DH}}(n) n^{-\sigma-\mathrm{i}\tau}\right| &\leq 2(1+\kappa) \sum_{m=0}^{\lfloor (M-1)/5 \rfloor} (5m+1)^{-1/2} \\ &\leq 2(1+\kappa)\Big(1 + \frac{2}{5}(\sqrt{\tau} - 1)\Big) \leq 2(1+\kappa)(0.46\tau^{1/2}) < 1.19\,\tau^{1/2},\end{aligned} \tag{A.1}$$

since $\sum_{r=1}^{5} |a_{\mathrm{DH}}(5m+r)| = 2(1+\kappa)$ and $\frac{3}{5} + \frac{2}{5}\sqrt{\tau} \leq 0.46\sqrt{\tau}$ for $\tau \geq 100$. Summing the tail $n \geq M+1$ by parts yields the remainder $-S(M)(M+1)^{-\sigma-\mathrm{i}\tau}$ plus $(\sigma+\mathrm{i}\tau)\int_{M+1}^{\infty} S(\lfloor x \rfloor) x^{-\sigma-1-\mathrm{i}\tau}\,\mathrm{d}x$. Since $M+1 = \lfloor \tau \rfloor + 1 > \tau \geq 100$, for $\sigma \leq 19\tau^{1/2}/20$ we have $|\sigma+\mathrm{i}\tau| \leq \tau\sqrt{1 + \frac{361}{400\tau}} \leq 1.005\,\tau$ and $\int_{M+1}^{\infty} x^{-3/2}\,\mathrm{d}x = 2(M+1)^{-1/2} < 2\tau^{-1/2}$, so the tail is at most $(1+\kappa)\tau^{-1/2} + 1.005(1+\kappa)\cdot 2\tau^{1/2} < 2.71\,\tau^{1/2}$, giving $|\mathcal{D}_{\mathrm{DH}}(\sigma+\mathrm{i}\tau)| < (1.19+2.71)\tau^{1/2} = 3.90\,\tau^{1/2} < 4\tau^{1/2}$. When $\sigma > 19\tau^{1/2}/20 \geq \frac{19}{2}$, $|\mathcal{D}_{\mathrm{DH}}(\sigma+\mathrm{i}\tau)| \leq \zeta(19/2) < 1.002 < 4\tau^{1/2}$. Finally, for $u \geq \frac{5}{4}$ ($\sigma = \frac{1}{2} + u \geq \frac{7}{4}$), monotonicity in σ and Hurwitz evaluation at $\sigma = \frac{7}{4}$ give $5^{-7/4}[\zeta(\frac{7}{4}, \frac{1}{5}) - 5^{7/4} + \kappa(\zeta(\frac{7}{4}, \frac{2}{5}) + \zeta(\frac{7}{4}, \frac{3}{5})) + \zeta(\frac{7}{4}, \frac{4}{5})] = 0.41770137\ldots < \frac{1}{2}$ for $\sum_{n=2}^{\infty} |a_{\mathrm{DH}}(n)| n^{-7/4}$, whence $\operatorname{Re}\mathcal{D}_{\mathrm{DH}} > 1 - \frac{1}{2} = \frac{1}{2}$ and $|\mathcal{D}_{\mathrm{DH}}| < 1 + \frac{1}{2} = \frac{3}{2}$. □

Completion of Binet and tail bounds in Proposition 4.3. Write $z_0 := \frac{3}{4} + \frac{\mathrm{i}\tilde{\gamma}}{2}$, so $|z_0| > \frac{\tilde{\gamma}}{2} \geq 200$. By Binet's second integral formula, $\log\Gamma(z_0 + \frac{u}{2}) - \log\Gamma(z_0) - \frac{u}{2}\log z_0 = \int_0^{u/2} [\psi(z_0 + w) - \log(z_0 + w) + \log(1 + \frac{w}{z_0})]\,\mathrm{d}w$. Since $|\psi(z) - \log z + \frac{1}{2z}| \leq \frac{1}{6|z|^2}$ on $\operatorname{Re} z > 0$ and $|\log(1 + \frac{w}{z_0})| \leq \frac{w}{|z_0|}$ for all $w \geq 0$, we obtain $|\log\frac{\Gamma(z_0+u/2)}{\Gamma(z_0)} - \frac{u}{2}\log z_0| \leq \frac{u^2+3u}{8|z_0|}$. Combined with $|\frac{u}{2}(\log z_0 - \log\frac{\mathrm{i}\tilde{\gamma}}{2})| = \frac{u}{2}|\log(1 - \frac{3\mathrm{i}}{2\tilde{\gamma}})| \leq \frac{3u}{4\tilde{\gamma}}$, this gives $|E_{\mathrm{St}}(u, \tilde{\gamma})| \leq \frac{u^2+6u}{4\tilde{\gamma}}$ for all $u \geq 0$. Set $X := \log\frac{5\tilde{\gamma}}{2\pi}$. By (4.1), $\frac{t}{4}X > \Delta + 5 + 6\sqrt{t} + 2t$ and $\tilde{\gamma} > \max(64t^3, \mathrm{e}^X) \geq (64t^3)^{2/3}(\mathrm{e}^X)^{1/3} = 16t^2\mathrm{e}^{X/3}$. Since $u_0 + 3\sqrt{t} = \frac{t}{4}X + \eta + 3\sqrt{t} \in (5, \frac{t}{2}X)$, on $u \in [0, u_0 + 3\sqrt{t}]$ we have $|E_{\mathrm{St}}(u, \tilde{\gamma})| < \frac{\frac{11}{20}t^2X^2}{64t^2\mathrm{e}^{X/3}} = \frac{11}{1280}X^2\mathrm{e}^{-X/3} \leq \frac{11}{1280}\cdot 36\mathrm{e}^{-2} < 0.042 < \frac{1}{10}$. On the right tail $u \geq u_0 + 3\sqrt{t}$, $\frac{u^2+6u}{4\tilde{\gamma}} \leq \frac{(u_0+3)^2+(u-u_0)^2}{\tilde{\gamma}} < \frac{1}{10} + \frac{(u-u_0)^2}{2t}$ (as $\tilde{\gamma} > 400 + 64t^3 > 2t$), so Lemma 4.2 bounds the right tail by $\frac{3}{2}\mathrm{e}^{1/10}\int_{3\sqrt{t}}^{\infty} \mathrm{e}^{-y^2/(2t)}\,\mathrm{d}y < 0.01\sqrt{\pi t}$. Splitting the left tail at $u = \frac{5}{4}$, on $[\frac{5}{4}, u_0 - 3\sqrt{t}]$ Lemma 4.2 gives $|\mathcal{D}_{\mathrm{DH}}| < \frac{3}{2}$ and $|E_{\mathrm{St}}| < \frac{1}{10}$, bounded by $\frac{3}{2}\mathrm{e}^{1/10}\int_{3\sqrt{t}}^{\infty} \mathrm{e}^{-y^2/t}\,\mathrm{d}y < 10^{-4}\sqrt{\pi t}$. On $[0, \frac{5}{4}]$ and

on $\mathcal{V}_\theta = \{\mathrm{i}y : 0 \le y \le \theta\}$, we have $u_0 - \frac{5}{4} > \max(\frac{15}{4} + 6\sqrt{t} + 2t,\ \frac{1}{6}tX)$, so splitting $(u_0 - \frac{5}{4})^2 = (\frac{17}{20} + \frac{3}{20})(u_0 - \frac{5}{4})^2$ gives $\frac{(u_0 - 5/4)^2}{t} > \frac{17}{32}X + \frac{2.1}{t} + \frac{6.75}{\sqrt{t}} + 5.4 + 0.6t$. Combining $|\mathcal{D}_{\mathrm{DH}}| \le 4\tilde{\gamma}^{1/2} < 4.52\mathrm{e}^{X/2}$ with $|\Gamma_{\mathrm{DH}}(\frac{1}{2} + \mathrm{i}(\gamma + y))/\Gamma_{\mathrm{DH}}(\frac{1}{2} + \mathrm{i}\tilde{\gamma})| \le \mathrm{e}^{\frac{\pi}{4}(\theta - y) + 0.01}$ and $\frac{\pi^2 t}{64} < 0.16t$ bounds the $[0, \frac{5}{4}]$ integral by $5\tilde{\gamma}^{1/2}\mathrm{e}^{0.01 - (u_0 - 5/4)^2/t} < 10^{-3}\sqrt{\pi t}$, $|\mathcal{E}_{\mathcal{V}}| \le \frac{\pi t}{2}\tilde{\gamma}^{1/2}\mathrm{e}^{0.01 - u_0^2/t + \pi^2 t/64} < 10^{-3}\sqrt{\pi t}$, and $|\xi_{\mathrm{DH}}(\frac{1}{2} + \mathrm{i}\gamma)/(\sqrt{\pi t}\, A_t(\gamma, 0))| \le 4\tilde{\gamma}^{1/2}\mathrm{e}^{0.01 - u_0^2/t + \pi^2 t/64}/(0.29\sqrt{\pi t}) < 10^{-2}$. Finally, for $\partial_\eta A_t(\gamma, \eta)$, integrating $\frac{2|u - u_0|}{t}\mathrm{e}^{-(u - u_0)^2/t}$ over $[\frac{5}{4}, u_0 + 3\sqrt{t}]$ contributes at most $3\mathrm{e}^{1/10} < \frac{1.88}{\sqrt{t}}\sqrt{\pi t}$ (with $< \frac{0.06}{\sqrt{t}}\sqrt{\pi t}$ from the remaining pieces via $\mathrm{e}^{-X/32}$), and dividing by $0.29\sqrt{\pi t}$ gives $< \frac{6.7}{\sqrt{t}} < \frac{\Delta + 5}{t} + \frac{6}{\sqrt{t}} + 2 < \frac{1}{4}X = \frac{1}{4}\log\frac{5\tilde{\gamma}}{2\pi}$, establishing (4.3) and (4.4). □

Appendix B. Ball-Arithmetic Truncation Enclosures

Proof of Lemma 5.1 and Cauchy bound in Theorem 3.2. For $n \ge N \ge 25$ and $u \in [0, U]$, we have $\mathrm{e}^{4u} \ge 1$ and $n^2 \ge Nn$, so $\mathrm{e}^{-\pi n^2 \mathrm{e}^{4u}/5} \le r_N^n$ with $r_N = \mathrm{e}^{-\pi N/5} < 10^{-6}$. Since $|a_{\mathrm{DH}}(n)| \le 1$ ($\kappa < 0.285 < 1$), summing $\sum_{n=N}^{\infty} n\, r_N^n = \frac{r_N^N(N - (N-1)r_N)}{(1 - r_N)^2}$ and using $|\partial_z^k \cos(zu)| \le u^k \mathrm{e}^{|y|u} \le U^k \mathrm{e}^{|y|U}$ over $[0, U]$ gives $E_1(U, N, k, |y|)$. For $u \ge U \ge 2$, $\mathrm{e}^{4u} \ge \mathrm{e}^8 > 2980$, so $\sum_{n=1}^{\infty} n\, \mathrm{e}^{-\pi n^2 \mathrm{e}^{4u}/5} \le 2\mathrm{e}^{-\pi \mathrm{e}^{4u}/5}$; moreover, for $0 \le t \le \frac{1}{4}$ and $|y| \le 1$, $tu^2 + (3 + |y|)u + k\log u \le \frac{u^2}{4} + 6u \le \frac{\pi}{10}\mathrm{e}^{4u} - 4u$ for all $u \ge 2$. Hence the integrand on $[U, \infty)$ is bounded by $16\mathrm{e}^{-4u - \frac{\pi}{10}\mathrm{e}^{4u}} \le \frac{40}{\pi}\mathrm{e}^{-4U} \cdot \frac{2\pi}{5}\mathrm{e}^{4u}\mathrm{e}^{-\frac{\pi}{10}\mathrm{e}^{4u}}$, which integrates explicitly to $E_2(U)$. Bounding $\sum_{n=1}^{N-1} n|a_{\mathrm{DH}}(n)|\, \mathrm{e}^{-\pi n^2 \mathrm{e}^{4u}/5} \le \frac{N(N-1)}{2}$ and $|u^2 \cos(zu)| \le U^2 \cosh(U y_{\max})$ on $[0, U]$ yields (5.2). Finally, in Theorem 3.2, $\tilde{\xi}_{\mathrm{DH}}(s(z_0) + w)$ is entire in w; covering $|w| = 4$ by 512 overlapping complex interval arcs W_j in 384-bit `FLINT/Arb` certifies $M_4 := \max_{1 \le j \le 512} \sup_{w \in W_j} |\tilde{\xi}_{\mathrm{DH}}(s(z_0) + w)| \le 6.58 \times 10^7$, so Cauchy's estimate bounds the Taylor remainder on $|u| \le h \le \frac{311}{500}$ by $|R_{100}(u)| \le M_4 \sum_{k=100}^{\infty} (h/4)^k = M_4 \frac{(h/4)^{100}}{1 - h/4} < 10^{-72}$. □

Use of AI and computational verification

Portions of the literature exploration, asymptotic stress-testing, manuscript drafting, and `FLINT/Arb` verification scripting were conducted with the assistance of an AI research assistant (**Gemini**). Every analytical inequality, contour deformation, Möbius inversion identity, and interval-arithmetic enclosure reported in this paper was independently checked against symbolic and arbitrary-precision ball-arithmetic computations; all proof certificates and vector figures are deterministically reproduced by the open-source script `scripts/verify_paper15_debruijn_newman_dh.py` using `python-flint` [13].

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